

Lecture Notes on Complex Numbers

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Chapter 1

Complex Numbers

1.1 Sums and Products

The complex numbers can be defined as ordered pairs of real numbers (x, y) subject to specific operations of addition and multiplication. We identify the set of complex numbers $\mathbb{C}$ with the xy -plane (the complex plane).

Definition 1.1.1 (Complex Number). A complex number z is defined as an ordered pair of real numbers:

$$z = (x, y)$$

where x is the *real part* of z , denoted $\operatorname{Re}(z)$, and y is the *imaginary part* of z , denoted $\operatorname{Im}(z)$.

Two complex numbers $z_1 = (x_1, y_1)$ and $z_2 = (x_2, y_2)$ are equal if and only if their real parts are equal and their imaginary parts are equal.

Definition 1.1.2 (Algebraic Operations). Let $z_1 = (x_1, y_1)$ and $z_2 = (x_2, y_2)$. We define sum and product as:

$$\begin{aligned} z_1 + z_2 &= (x_1 + x_2, y_1 + y_2) \\ z_1 z_2 &= (x_1 x_2 - y_1 y_2, y_1 x_2 + x_1 y_2) \end{aligned}$$

Remark 1.1.1. The set of complex numbers contains the set of real numbers as a subset. We identify the real number x with the pair $(x, 0)$. Under this identification, operations correspond to standard real arithmetic.

1.1.1 The Imaginary Unit

It is customary to denote the number $(0, 1)$ by i . Using the definition of multiplication:

$$i^2 = (0, 1)(0, 1) = (0 \cdot 0 - 1 \cdot 1, 1 \cdot 0 + 0 \cdot 1) = (-1, 0) = -1$$

Thus, we can write any complex number $z = (x, y)$ as:

$$z = x + iy$$

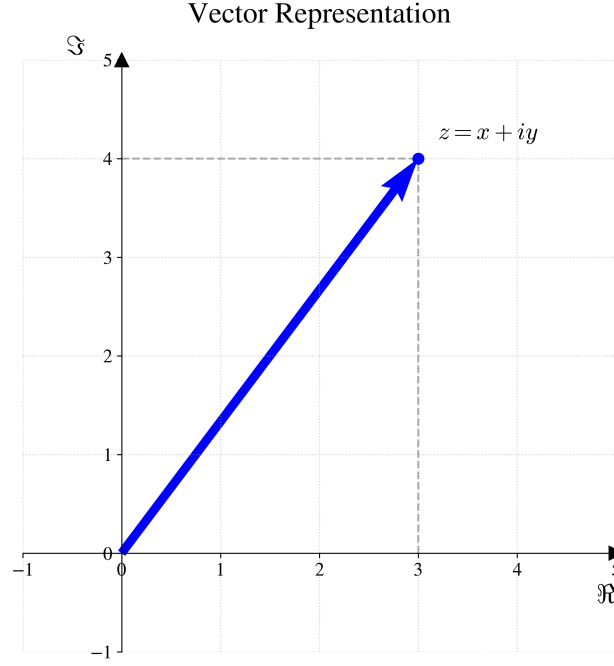


Figure 1.1: Vector representation of a complex number $z = x + iy$. The point (x, y) corresponds to the complex number in the plane.

1.2 Basic Algebraic Properties

The set $\mathbb{C}$ forms a **field** under addition and multiplication.

Theorem 1.2.1 (Field Properties). *For any $z_1, z_2, z_3 \in \mathbb{C}$, the following laws hold:*

1. **Commutative laws:** $z_1 + z_2 = z_2 + z_1$ and $z_1 z_2 = z_2 z_1$.
2. **Associative laws:** $(z_1 + z_2) + z_3 = z_1 + (z_2 + z_3)$ and $(z_1 z_2) z_3 = z_1 (z_2 z_3)$.
3. **Distributive law:** $z(z_1 + z_2) = z z_1 + z z_2$.
4. **Identities:** Additive identity is $0 = (0, 0)$. Multiplicative identity is $1 = (1, 0)$.
5. **Inverses:** For every z , there is an additive inverse $-z$. For every $z \neq 0$, there is a multiplicative inverse z^{-1} .

Example 1.2.1. Find the multiplicative inverse of $z = x + iy$ assuming $z \neq 0$. We seek $u + iv$ such that $(x + iy)(u + iv) = 1$. Solving the system of linear equations yields:

$$z^{-1} = \left(\frac{x}{x^2 + y^2}, \frac{-y}{x^2 + y^2} \right)$$

1.3 Vectors and Moduli

A complex number $z = x + iy$ is naturally associated with a position vector in the plane starting from the origin and terminating at (x, y) .

Definition 1.3.1 (Modulus). The modulus (or absolute value) of $z = x + iy$ is the non-negative real number:

$$|z| = \sqrt{x^2 + y^2}$$

Geometrically, this represents the distance of the point z from the origin.

Lemma 1.3.1 (Distance between points). *The distance between two complex numbers z_1 and z_2 is given by $|z_1 - z_2|$.*

Exercise 1.3.1. Prove that $|z|^2 = (\operatorname{Re}(z))^2 + (\operatorname{Im}(z))^2$.

1.4 The Triangle Inequality

One of the most crucial inequalities in analysis is the Triangle Inequality, which states that the length of a side of a triangle is less than or equal to the sum of the lengths of the other two sides.

Theorem 1.4.1 (Triangle Inequality). *For any $z_1, z_2 \in \mathbb{C}$:*

$$|z_1 + z_2| \leq |z_1| + |z_2|$$

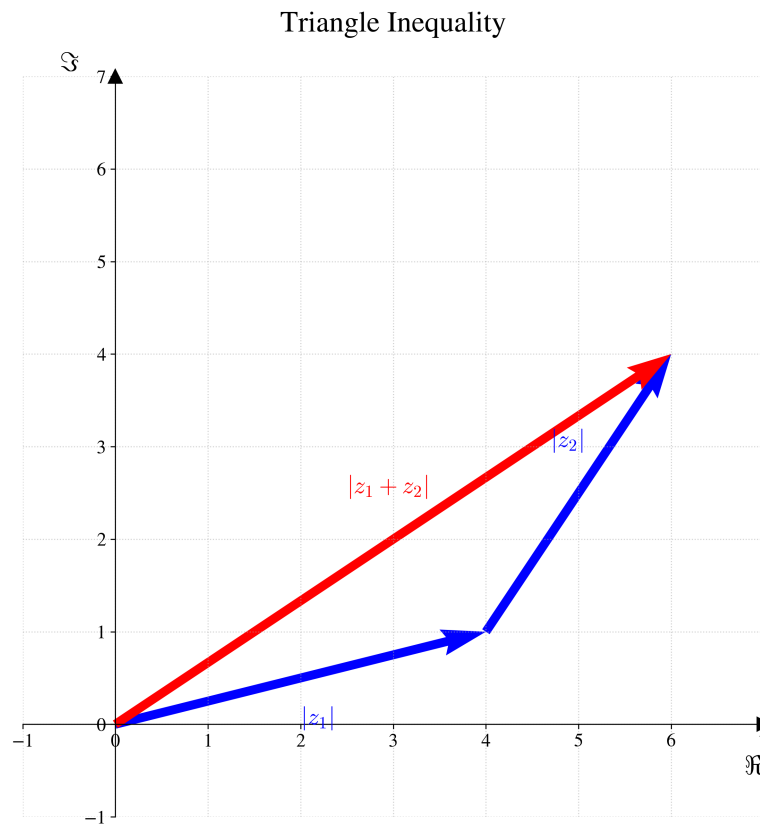


Figure 1.2: Geometric representation of the Triangle Inequality for complex numbers z_1 and z_2 . The length of the vector $z_1 + z_2$ is less than or equal to the sum of the lengths of z_1 and z_2 .

Corollary 1.4.2. *It follows from the theorem that:*

$$|z_1 + z_2| \geq ||z_1| - |z_2||$$

See Figure 1.2 for a geometric illustration.

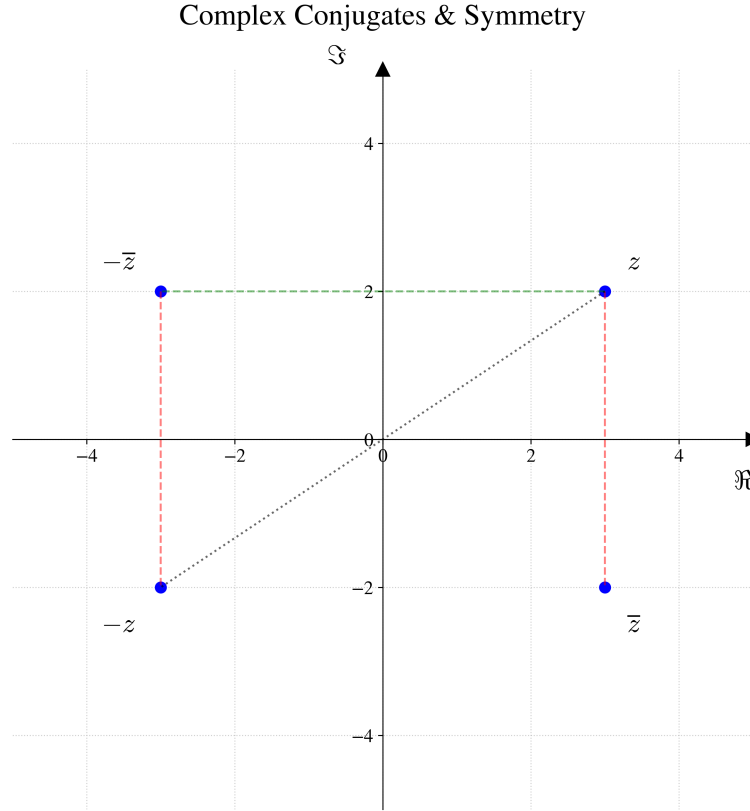


Figure 1.3: Geometric representation of the complex conjugate. The point $z = x + iy$ is reflected across the real axis to obtain $\bar{z} = x - iy$; the dashed line indicates the reflection. Analogously, $-z$ is the reflection across the origin, and $-\bar{z}$ is the reflection across the imaginary axis.

1.5 Complex Conjugates

Definition 1.5.1 (Conjugate). The complex conjugate of a number $z = x + iy$ is defined as:

$$\bar{z} = x - iy$$

Geometrically, $\bar{z}$ is the reflection of the point z across the real axis.

Theorem 1.5.1 (Properties of Conjugates). *For all $z, z_1, z_2 \in \mathbb{C}$:*

1. $\overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2$
2. $\overline{z_1 z_2} = \bar{z}_1 \bar{z}_2$
3. $|z|^2 = z \bar{z}$
4. $\operatorname{Re}(z) = \frac{z + \bar{z}}{2}$ and $\operatorname{Im}(z) = \frac{z - \bar{z}}{2i}$

1.6 Exponential Form

Using polar coordinates (r, θ) where $x = r \cos \theta$ and $y = r \sin \theta$, we can express complex numbers in exponential form.

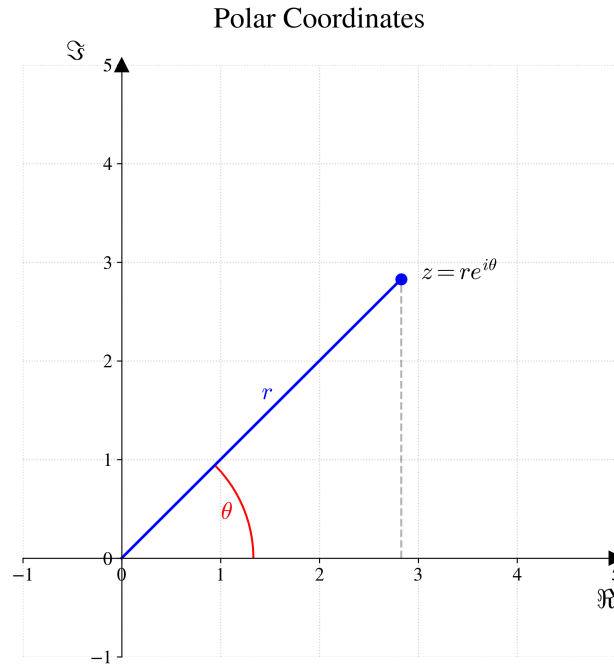


Figure 1.4: Exponential form of a complex number $z = re^{i\theta}$. The modulus r represents the distance from the origin, and the angle θ (argument) represents the direction from the positive real axis.

Definition 1.6.1 (Argument). The argument of z , denoted $\arg(z)$, is the set of angles θ such that $z = r(\cos \theta + i \sin \theta)$.

$$\theta = \arg(z) = \Theta + 2n\pi, \quad n \in \mathbb{Z}$$

where Θ is the **Principal Argument**, denoted $\text{Arg}(z)$, such that $-\pi < \Theta \leq \pi$.

Theorem 1.6.1 (Euler's Formula). For any real number θ ,

$$e^{i\theta} = \cos \theta + i \sin \theta$$

Thus, any non-zero complex number can be written as:

$$z = re^{i\theta}$$

1.7 Products and Powers in Exponential Form

Exponential form makes multiplication and division significantly easier geometrically.

Theorem 1.7.1 (Multiplication in Polar Form). If $z_1 = r_1 e^{i\theta_1}$ and $z_2 = r_2 e^{i\theta_2}$, then:

$$z_1 z_2 = r_1 r_2 e^{i(\theta_1 + \theta_2)}$$

$$\frac{z_1}{z_2} = \frac{r_1}{r_2} e^{i(\theta_1 - \theta_2)}$$

Remark 1.7.1. Multiplication by z scales the modulus by $|z|$ and rotates the vector by $\arg(z)$.

1.7.1 De Moivre's Formula

From the multiplication rule, we derive the formula for integer powers.

Theorem 1.7.2 (De Moivre's Formula). *For any integer n ,*

$$(\cos \theta + i \sin \theta)^n = \cos(n\theta) + i \sin(n\theta)$$

or equivalently, $(e^{i\theta})^n = e^{in\theta}$.

Example 1.7.1. Calculate $(1 + i)^{10}$. First, convert $1 + i$ to exponential form:

$$r = \sqrt{1^2 + 1^2} = \sqrt{2}, \quad \text{Arg}(z) = \frac{\pi}{4}$$

$$z = \sqrt{2}e^{i\pi/4}$$

$$z^{10} = (\sqrt{2})^{10}e^{i10\pi/4} = 2^5e^{i5\pi/2} = 32e^{i(\pi/2+2\pi)} = 32e^{i\pi/2} = 32i$$

1.8 Roots of Complex Numbers

Finding the n -th roots of a complex number z_0 involves solving $z^n = z_0$.

Theorem 1.8.1 (Roots). *The n distinct n -th roots of $z_0 = r_0e^{i\theta_0}$ are given by:*

$$z_k = \sqrt[n]{r_0} \exp\left(i \frac{\theta_0 + 2k\pi}{n}\right), \quad k = 0, 1, \dots, n-1$$

Geometrically, the roots lie on a circle of radius $\sqrt[n]{r_0}$ centered at the origin and form the vertices of a regular n -sided polygon.

1.9 Regions in the Complex Plane

To do calculus, we need topological definitions regarding sets of points in the plane.

Definition 1.9.1 (Epsilon Neighborhood). An ε -neighborhood of a point z_0 is the set of all points z such that:

$$|z - z_0| < \varepsilon$$

Definition 1.9.2 (Topological Concepts). The following definitions apply to sets of complex numbers:

- **Interior Point:** A point z_0 in a set S is an interior point if there exists some neighborhood of z_0 completely contained in S .
- **Open Set:** A set S is open if every point in S is an interior point.
- **Closed Set:** A set S is closed if it contains all its boundary points.
- **Domain:** An open, connected set.
- **Region:** A domain together with some, none, or all of its boundary points.

Exercise 1.9.1. Determine if the set $S = \{z \in \mathbb{C} \mid |z| < 1\}$ is open or closed.

Solution: It is open, since for each point z_0 in S , we can find define $\varepsilon = \frac{1}{3}\text{dist}(z_0, \partial S)$; then the set $\mathcal{U}_\varepsilon(z_0) = \{z \in \mathbb{C} \mid |z - z_0| < \varepsilon\}$ is an ε -neighborhood that lies entirely within S .

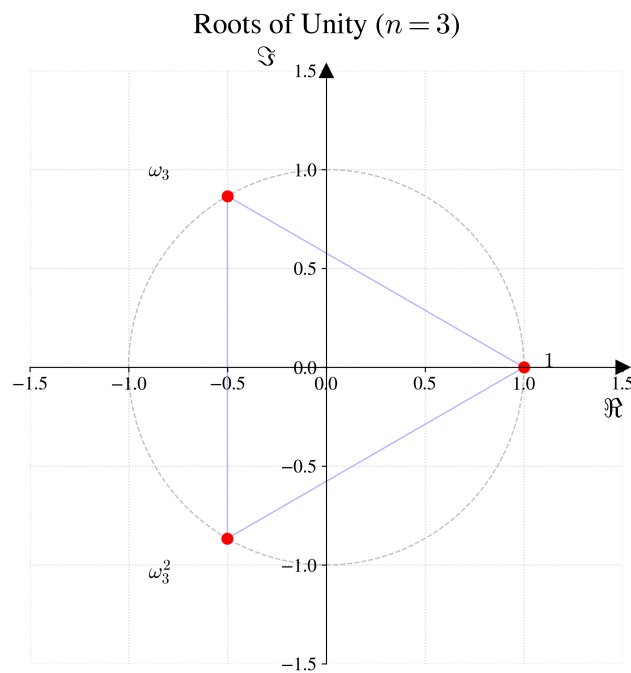


Figure 1.5: The n -th roots of a complex number z_0 are evenly spaced on a circle in the complex plane, forming the vertices of a regular n -sided polygon.

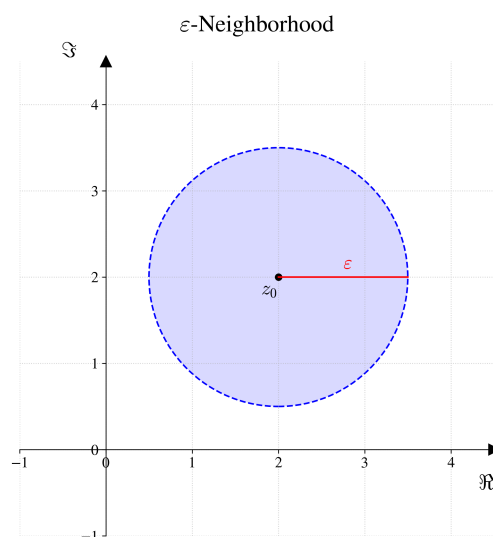


Figure 1.6: An ε -neighborhood around the point z_0 . The shaded area represents all points z such that $|z - z_0| < \varepsilon$.

1.10 Exercises

Exercise 1.10.1. Write the following complex numbers in the form $x + iy$:

1. $(5 - 2i) + (3 + 4i)$
2. $(2 + 3i)(4 - i)$
3. $\frac{1}{1+i}$
4. $\frac{3-2i}{-1+i}$

Exercise 1.10.2. Find $\operatorname{Re}(z)$ and $\operatorname{Im}(z)$ for the following:

1. $z = \frac{1}{i}$
2. $z = (1 - i)^3$
3. $z = \frac{2+i}{3-4i} + \frac{2-i}{5i}$

Exercise 1.10.3. Let $z_1 = 1 - i$ and $z_2 = -2 + 4i$. Compute:

1. $z_1^2 - 2z_1 + 3$
2. $|2z_1 - 3z_2|$
3. $\operatorname{Re}(z_1 z_2)$

Exercise 1.10.4. Verify that $(\sqrt{2} - i) - i(1 - \sqrt{2}i) = -2i$.

Exercise 1.10.5. Solve the following equation for z :

$$z(1 + i) = \bar{z} + (3 + 2i)$$

Hint: Let $z = x + iy$ and equate real and imaginary parts.

Exercise 1.10.6. Verify the following properties using $z = x + iy$:

1. $\overline{iz} = -i\bar{z}$
2. $\overline{\bar{z} + 3i} = z - 3i$
3. $|z|^2 = z\bar{z}$

Exercise 1.10.7. Prove that for any $z_1, z_2 \in \mathbb{C}$:

$$|z_1 + z_2|^2 + |z_1 - z_2|^2 = 2(|z_1|^2 + |z_2|^2)$$

This is known as the *Parallelogram Law*.

Exercise 1.10.8. Show that $|z_1 - z_2|^2 = |z_1|^2 + |z_2|^2 - 2\operatorname{Re}(z_1 \bar{z}_2)$.

Exercise 1.10.9. (*Hard*) If $|z| = 1$, prove that:

$$\left| \frac{z - w}{1 - \bar{w}z} \right| = 1$$

for any complex number w with $|w| \neq 1$.

Exercise 1.10.10. Write the following numbers in exponential form $re^{i\theta}$ ($-\pi < \theta \leq \pi$):

1. $1 - i$
2. $-2\sqrt{3} - 2i$
3. $-i$
4. $(1 - i\sqrt{3})^2$

Exercise 1.10.11. Find the Principal Argument, $\text{Arg}(z)$, for:

1. $z = \frac{-2}{1+i\sqrt{3}}$
2. $z = (\sqrt{3} - i)^6$

Exercise 1.10.12. Using exponential form, show that:

$$(-1 + i)^7 = -8(1 + i)$$

Exercise 1.10.13. Compute the following powers and write the result in the form $x + iy$:

1. $(1 + i)^{10}$
2. $\left(\frac{1-i\sqrt{3}}{2}\right)^{15}$
3. $(1 - i)^{-8}$

Exercise 1.10.14. Compute the value of $(1 + \cos \frac{\pi}{3} + i \sin \frac{\pi}{3})^6$.

Hint: Use De Moivre's formula.

Exercise 1.10.15. Find all distinct roots of the following equations and locate them in the complex plane:

1. $z^3 = 1$
2. $z^2 + 2i = 0$
3. $z^4 = -16$

Exercise 1.10.16. Find the four values of $(-1)^{1/4}$.

Exercise 1.10.17. Solve the quadratic equation $z^2 + z + 1 = 0$ for z .

Exercise 1.10.18. Sketch the set of points determined by the condition:

1. $|z - 1 + i| = 1$
2. $|z + i| \leq 3$
3. $\text{Re}(\bar{z} - i) = 2$
4. $|2z - i| = 4$

Exercise 1.10.19. Describe the set of points z such that $\text{Im}(z^2) > 0$.

Exercise 1.10.20. Prove that if $1 + z + z^2 + \cdots + z^n = 0$, then z is a root of unity, i.e. $z = e^{\frac{2\pi ik}{n+1}}$ for some integer k .

Exercise 1.10.21. Establish the Lagrange trigonometric identity:

$$1 + \cos \theta + \cos 2\theta + \cdots + \cos n\theta = \frac{1}{2} + \frac{\sin[(n + 1/2)\theta]}{2 \sin(\theta/2)}$$

where $0 < \theta < 2\pi$.

Chapter 2

Analytic Functions

2.1 Functions of a Complex Variable

A complex function f is a rule that assigns to each complex number z in a set $S \subseteq \mathbb{C}$ a complex number w . We write:

$$w = f(z)$$

Here, S is the **domain of definition** of f . Since $z = x + iy$, the function $f(z)$ can be decomposed into its real and imaginary parts:

$$f(z) = u(x, y) + iv(x, y)$$

where u and v are real-valued functions of the real variables x and y .

Example 2.1.1. Consider $f(z) = z^2$.

$$f(x + iy) = (x + iy)^2 = x^2 - y^2 + i(2xy)$$

Thus, $u(x, y) = x^2 - y^2$ and $v(x, y) = 2xy$.

We cannot always draw complex functions, actually we usually cannot draw them at all since they map from a 2D space (the complex plane) to another 2D space (another complex plane), resulting in a 4D space which is not directly visualizable. However, we can analyze how certain sets in the z -plane are transformed into sets in the w -plane. This is often done by examining the images of lines, circles, and other simple shapes under the mapping defined by f . For this reason, we usually think about complex function as mappings from the z -plane to the w -plane. Therefore, we describe this function by means of their **geometric** properties. For example

1. We say that the map $w = z + 1 + i$ is a translation of the z -plane by the vector $(1, 1)$.
2. The map $w = \bar{z}$ is a reflection about the real axis.
3. The map $w = iz$ is a rotation of the z -plane by 90° counterclockwise about the origin.

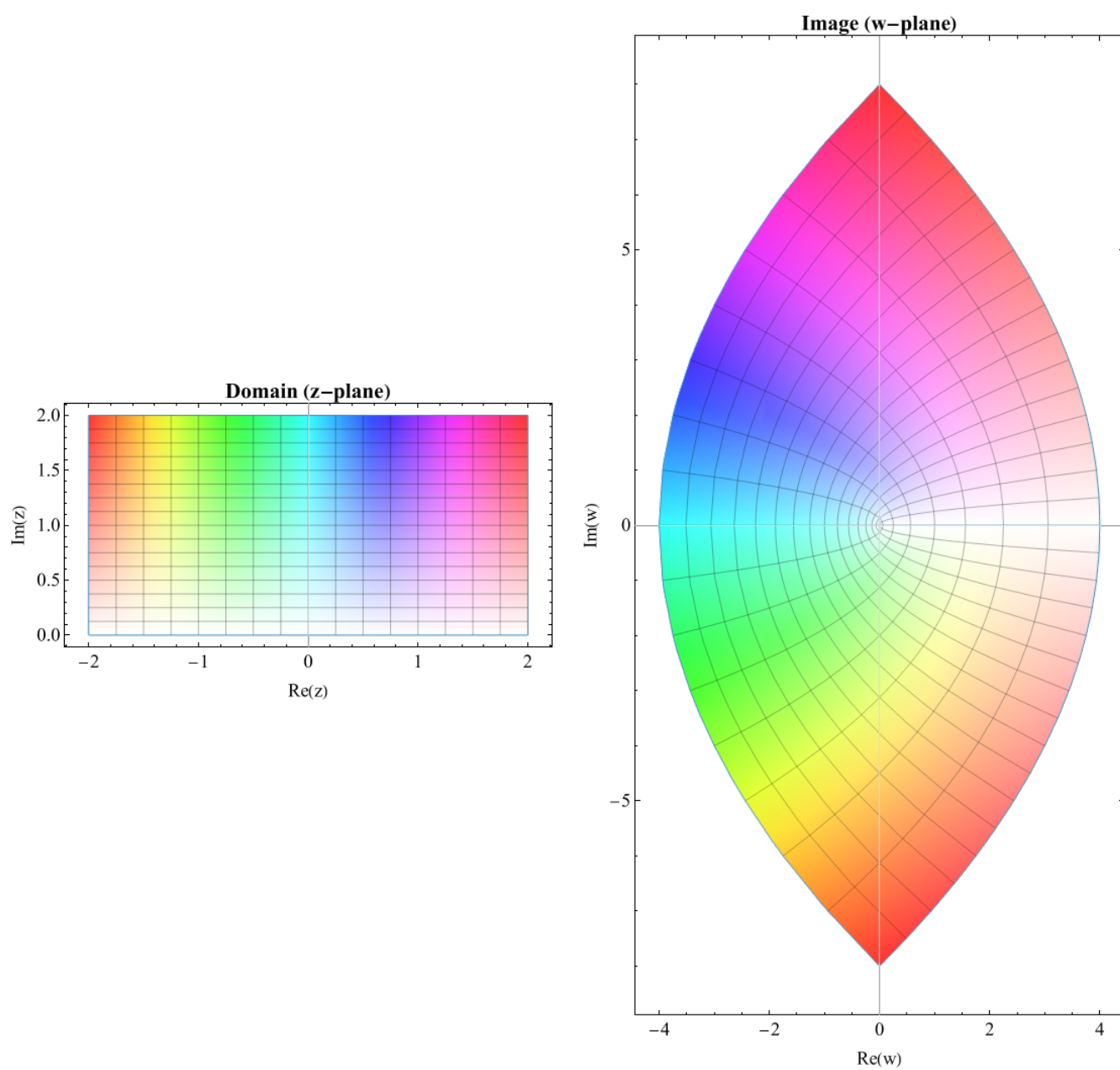
Mapping $w = z^2$ 

Figure 2.1: Mapping of a complex function from the z -plane to the w -plane.

2.1.1 Real valued functions

A special case of complex functions are those that map complex numbers to real numbers, i.e., $f : \mathbb{C} \rightarrow \mathbb{R}$. In this case, the imaginary part $v(x, y)$ is identically zero, this complex functions becomes identical to a real-valued function of two real variables:

$$f(z) = u(x, y) + i \cdot 0 = u(x, y),$$

therefore we can draw their graphs in the usual 3D space with axes x, y, u . Another way of getting information from complex valued functions is to look at the graph of their real and imaginary parts separately as surfaces in 3D space.

2.1.2 Example: The Joukowski Transformation

Let us examine the mapping defined by the function:

$$w = f(z) = z + \frac{1}{z}$$

This function is historically significant in aerodynamics for mapping circles into airfoil shapes. To understand the geometry of this mapping, it is most convenient to use polar coordinates $z = re^{i\theta}$. Substituting into the function:

$$w = re^{i\theta} + \frac{1}{re^{i\theta}} = r(\cos \theta + i \sin \theta) + \frac{1}{r}(\cos \theta - i \sin \theta)$$

Separating the real and imaginary parts $w = u + iv$, we obtain:

$$u = \left(r + \frac{1}{r}\right) \cos \theta, \quad v = \left(r - \frac{1}{r}\right) \sin \theta$$

Mapping of the Unit Circle ($r = 1$)

Consider the unit circle $|z| = 1$. Here $r = 1$, so the equations become:

$$u = 2 \cos \theta, \quad v = 0$$

As θ varies from 0 to 2π , u varies from 2 to -2 and back to 2. Thus, the unit circle in the z -plane is mapped onto the line segment $[-2, 2]$ on the real axis of the w -plane.

Mapping of Circles ($r = c > 1$)

Now consider a circle $|z| = c$ where $c > 1$. The equations for u and v are:

$$u = A \cos \theta, \quad v = B \sin \theta$$

where $A = c + \frac{1}{c}$ and $B = c - \frac{1}{c}$. Using the identity $\cos^2 \theta + \sin^2 \theta = 1$, we can eliminate θ :

$$\frac{u^2}{A^2} + \frac{v^2}{B^2} = 1$$

This is the equation of an ellipse with semi-major axis A and semi-minor axis B . Note that the foci of this ellipse are located at $\pm\sqrt{A^2 - B^2} = \pm\sqrt{4} = \pm 2$, which are the endpoints of the slit created by the unit circle mapping.

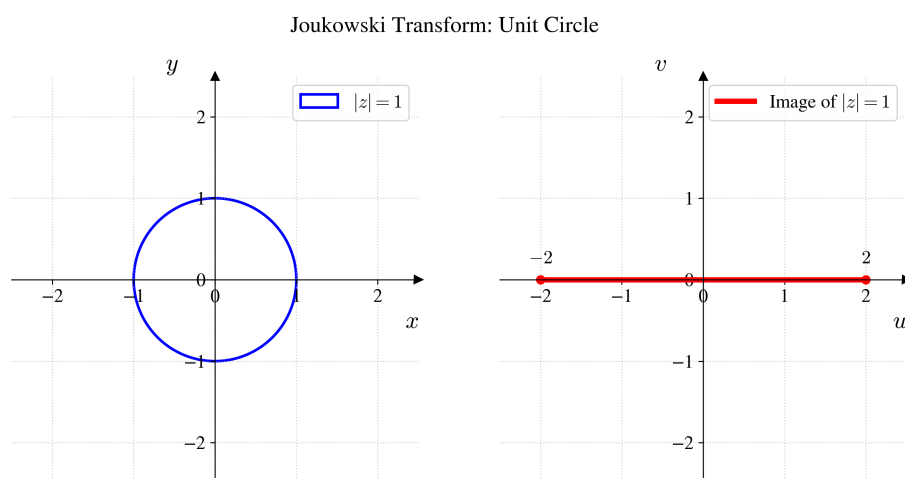
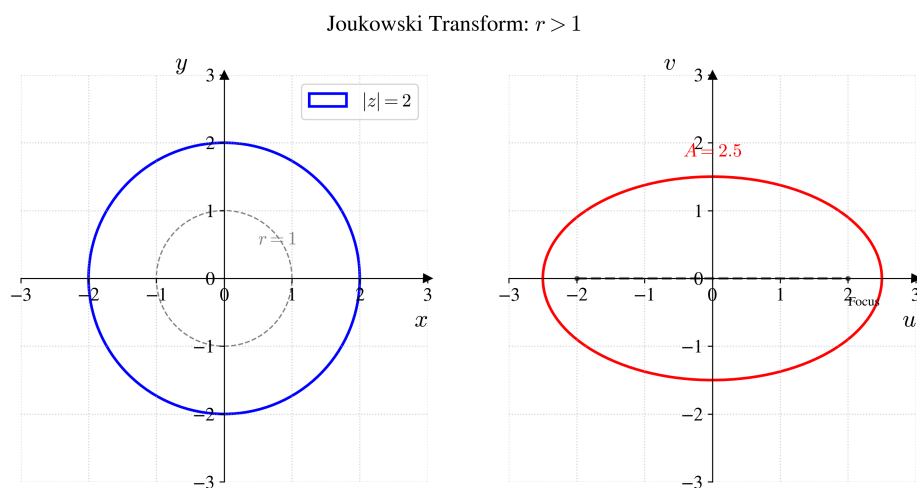


Figure 2.2: Mapping of the unit circle under the Joukowski transformation.

Figure 2.3: Mapping of the circle $|z| = c$ under the Joukowski transformation.

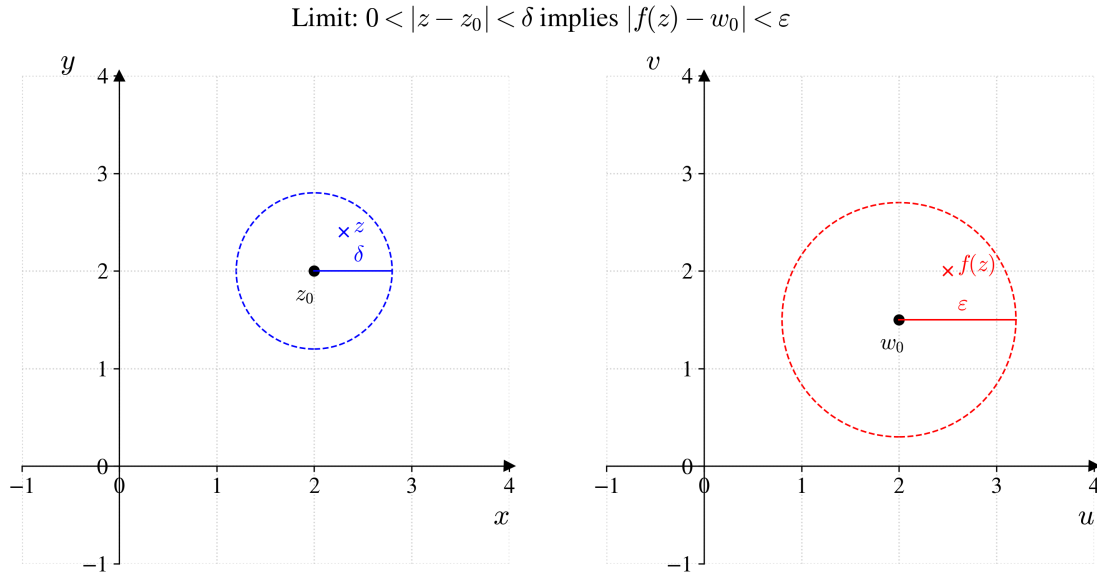


Figure 2.4: Illustration of the limit definition in the complex plane.

2.2 Limits and Continuity

The concept of a limit in the complex plane is similar to that in real calculus, but since the neighborhood is two-dimensional, z can approach z_0 from any direction.

Definition 2.2.1 (Limit). Let f be defined in a neighborhood of z_0 , except possibly at z_0 itself. We say that the limit of $f(z)$ as z approaches z_0 is w_0 , written

$$\lim_{z \rightarrow z_0} f(z) = w_0$$

if, for every $\varepsilon > 0$, there exists a $\delta > 0$ such that:

$$|f(z) - w_0| < \varepsilon \quad \text{whenever} \quad 0 < |z - z_0| < \delta$$

Theorem 2.2.1 (Continuity). A function f is continuous at z_0 if:

1. $f(z_0)$ is defined,
2. $\lim_{z \rightarrow z_0} f(z)$ exists, and
3. $\lim_{z \rightarrow z_0} f(z) = f(z_0)$.

2.3 Derivatives

Differentiation is the cornerstone of complex analysis. The definition is formally identical to the real case, but the implications are far more profound.

Definition 2.3.1 (Derivative). Let f be defined in a neighborhood of z_0 . The derivative of f at z_0 , denoted $f'(z_0)$, is:

$$f'(z_0) = \lim_{\Delta z \rightarrow 0} \frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z}$$

provided this limit exists.

If the limit exists, f is said to be **differentiable** at z_0 . The approach $\Delta z \rightarrow 0$ can occur along the real axis ($\Delta x \rightarrow 0$), the imaginary axis ($i\Delta y \rightarrow 0$), or any other path. The limit must be independent of the path.

Remark 2.3.1. Standard differentiation rules (sum, product, quotient, chain rule) from calculus apply to complex derivatives.

2.3.1 Examples of Differentiability

To illustrate the definition, let us look at one function that is differentiable everywhere and one that is not differentiable anywhere.

Example 2.3.1 (Differentiable Function). Consider the function $f(z) = z^2$. Using the definition of the derivative:

$$\begin{aligned} f'(z) &= \lim_{\Delta z \rightarrow 0} \frac{(z + \Delta z)^2 - z^2}{\Delta z} \\ &= \lim_{\Delta z \rightarrow 0} \frac{z^2 + 2z\Delta z + (\Delta z)^2 - z^2}{\Delta z} \\ &= \lim_{\Delta z \rightarrow 0} (2z + \Delta z) = 2z \end{aligned}$$

Since the limit $2z$ exists and is independent of the path taken by Δz , $f(z) = z^2$ is differentiable everywhere (it is an entire function).

Example 2.3.2 (Non-Differentiable Function). Consider the function $f(z) = \bar{z} = x - iy$. We attempt to find the limit of the difference quotient:

$$\lim_{\Delta z \rightarrow 0} \frac{\overline{z + \Delta z} - \bar{z}}{\Delta z} = \lim_{\Delta z \rightarrow 0} \frac{\overline{\Delta z}}{\Delta z}$$

Let $\Delta z = \Delta x + i\Delta y$.

- **Path 1 (Along real axis):** Set $\Delta y = 0$, so $\Delta z = \Delta x$.

$$\lim_{\Delta x \rightarrow 0} \frac{\Delta x}{\Delta x} = 1$$

- **Path 2 (Along imaginary axis):** Set $\Delta x = 0$, so $\Delta z = i\Delta y$.

$$\lim_{\Delta y \rightarrow 0} \frac{-i\Delta y}{i\Delta y} = -1$$

Since the limits along these two paths are different ($1 \neq -1$), the derivative $f'(z)$ does not exist anywhere.

2.4 Cauchy-Riemann Equations

Since $f(z) = u(x, y) + iv(x, y)$, there is a necessary relationship between the partial derivatives of u and v for $f'(z)$ to exist.

Theorem 2.4.1 (Cauchy-Riemann Equations). *Suppose $f(z) = u(x, y) + iv(x, y)$ is differentiable at a point $z_0 = x_0 + iy_0$. Then the partial derivatives of u and v exist at (x_0, y_0) and satisfy the **Cauchy-Riemann equations**:*

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

Furthermore, the derivative is given by:

$$f'(z_0) = u_x + iv_x$$

Proof. Assume $f(z) = u(x, y) + iv(x, y)$ is differentiable at a point z . Then the limit

$$f'(z) = \lim_{\Delta z \rightarrow 0} \frac{f(z + \Delta z) - f(z)}{\Delta z}$$

exists and is unique regardless of how Δz approaches 0.

Case 1: Approach along the horizontal axis ($\Delta y = 0$).

Let $\Delta z = \Delta x$. Then:

$$\begin{aligned} f'(z) &= \lim_{\Delta x \rightarrow 0} \frac{u(x + \Delta x, y) - u(x, y)}{\Delta x} + i \lim_{\Delta x \rightarrow 0} \frac{v(x + \Delta x, y) - v(x, y)}{\Delta x} \\ f'(z) &= u_x(x, y) + iv_x(x, y) \end{aligned} \tag{2.1}$$

Case 2: Approach along the vertical axis ($\Delta x = 0$).

Let $\Delta z = i\Delta y$. Then:

$$f'(z) = \lim_{\Delta y \rightarrow 0} \frac{u(x, y + \Delta y) - u(x, y)}{i\Delta y} + i \lim_{\Delta y \rightarrow 0} \frac{v(x, y + \Delta y) - v(x, y)}{i\Delta y}$$

Using $1/i = -i$, we get:

$$f'(z) = -iu_y(x, y) + v_x(x, y) \tag{2.2}$$

Equating the real and imaginary parts (2.1)-(2.2) gives the Cauchy-Riemann equations:

$$u_x = v_y \quad \text{and} \quad v_x = -u_y$$

□

Theorem 2.4.2 (Sufficient Conditions). *If the partial derivatives u_x, u_y, v_x, v_y exist, are continuous in a neighborhood of z_0 , and satisfy the Cauchy-Riemann equations at z_0 , then $f'(z_0)$ exists.*

2.4.1 Polar Form

Using $z = re^{i\theta}$, the Cauchy-Riemann equations in polar coordinates are:

$$ru_r = v_\theta, \quad u_\theta = -rv_r$$

and the derivative is given by:

$$f'(z) = e^{-i\theta}(u_r + iv_r)$$

Proof. Let $z = re^{i\theta}$ and $f(z) = u(r, \theta) + iv(r, \theta)$. Assuming f is differentiable at a non-zero point z , we can compute the partial derivatives with respect to r and θ using the chain rule.

Recall that $z = re^{i\theta}$.

1. Differentiate with respect to r :

$$\frac{\partial f}{\partial r} = f'(z) \frac{\partial z}{\partial r} = f'(z)e^{i\theta}$$

Also, directly from the components:

$$\frac{\partial f}{\partial r} = u_r + iv_r$$

Equating these implies:

$$f'(z) = e^{-i\theta}(u_r + iv_r) \quad (2.3)$$

2. Differentiate with respect to θ :

$$\frac{\partial f}{\partial \theta} = f'(z) \frac{\partial z}{\partial \theta} = f'(z)(ire^{i\theta}) = irf'(z)e^{i\theta}$$

Also, directly from the components:

$$\frac{\partial f}{\partial \theta} = u_\theta + iv_\theta$$

Solving for $f'(z)$:

$$f'(z) = \frac{1}{ire^{i\theta}}(u_\theta + iv_\theta) = \frac{e^{-i\theta}}{r}(v_\theta - iu_\theta) \quad (2.4)$$

Equating (2.3)-(2.4) gives:

$$u_r + iv_r = \frac{1}{r}v_\theta - i\frac{1}{r}u_\theta$$

Matching real and imaginary parts yields the polar form:

$$u_r = \frac{1}{r}v_\theta, \quad v_r = -\frac{1}{r}u_\theta$$

□

2.5 Analytic Functions

Definition 2.5.1 (Analyticity). A function f is **analytic** at a point z_0 if it is differentiable not only at z_0 but also at every point in some neighborhood of z_0 . A function is analytic in a domain D if it is analytic at every point in D .

Definition 2.5.2 (Entire Function). A function that is analytic at every point in the complex plane $\mathbb{C}$ is called an **entire function**. Examples include polynomials and e^z .

Theorem 2.5.1. If $f'(z) = 0$ everywhere in a domain D , then $f(z)$ is constant in D .

Definition 2.5.3 (Singular Point). A point z_0 is a **singular point** of a function f if f is not analytic at z_0 but is analytic at some point in every neighborhood of z_0 .

Example 2.5.1. The function $f(z) = \frac{1}{z}$ is analytic everywhere except at $z = 0$, which is a singular point.

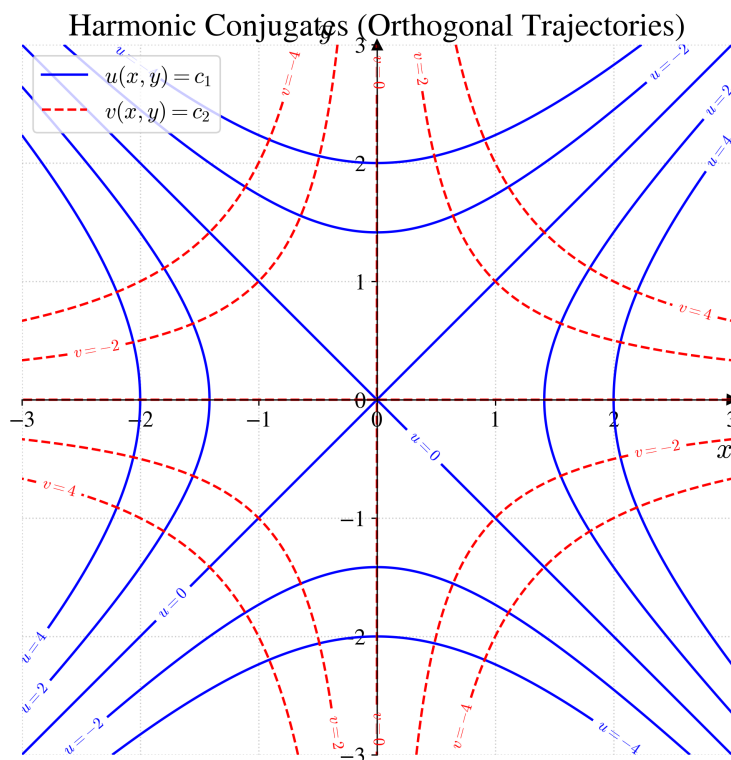


Figure 2.5: Harmonic functions and their conjugates.

2.6 Harmonic Functions

Analytic functions are closely related to the Laplace equation from physics and engineering.

Definition 2.6.1 (Harmonic Function). A real-valued function $H(x, y)$ is **harmonic** in a domain if it has continuous partial derivatives of the first and second order and satisfies Laplace's equation:

$$H_{xx}(x, y) + H_{yy}(x, y) = 0$$

Theorem 2.6.1. If $f(z) = u(x, y) + iv(x, y)$ is analytic in a domain D , then its component functions u and v are harmonic in D .

Definition 2.6.2 (Harmonic Conjugate). If u and v are harmonic in a domain D and their first-order partial derivatives satisfy the Cauchy-Riemann equations, then v is called a **harmonic conjugate** of u .

2.7 Exercises

Exercise 2.7.1. Determine the domain of definition for the following functions:

1. $f(z) = \frac{1}{z^2+1}$
2. $f(z) = \text{Arg}\left(\frac{1}{z}\right)$

Exercise 2.7.2. Using the definition of the derivative (limits), show that the function $f(z) = \text{Re}(z)$ is nowhere differentiable.

Exercise 2.7.3. Use the Cauchy-Riemann equations to determine where the following functions are differentiable and where they are analytic:

1. $f(z) = \bar{z}$
2. $f(z) = e^{-x}(\cos y - i \sin y)$
3. $f(z) = z^2 + z$

Exercise 2.7.4. Show that the function $u(x, y) = 2x(1 - y)$ is harmonic. Find a function $v(x, y)$ such that $f(z) = u + iv$ is analytic. Express $f(z)$ in terms of z .

Exercise 2.7.5. Prove that if $f(z)$ is analytic in a domain D and $|f(z)|$ is constant in D , then $f(z)$ itself must be constant in D .

Exercise 2.7.6. Using the polar form of the Cauchy-Riemann equations, show that the function $f(z) = \sqrt{r}e^{i\theta/2}$ (where $r > 0$ and $-\pi < \theta < \pi$) is analytic in its domain. Compute $f'(z)$.

Exercise 2.7.7. Consider the function $f(z) = |z|^2$.

1. Write $f(z)$ in terms of x and y (i.e., find $u(x, y)$ and $v(x, y)$).
2. Apply the Cauchy-Riemann equations to determine the set of points where $f'(z)$ exists.
3. Is $f(z)$ analytic at any point? Explain the difference between your answer here and the result in part 2.

Chapter 3

Elementary Functions

3.1 The Exponential Function

We extend the real exponential function e^x to the complex plane in a way that preserves its fundamental properties (derivative, addition laws).

Definition 3.1.1 (Complex Exponential). Let $z = x + iy$. The exponential function e^z (or $\exp z$) is defined as:

$$e^z = e^x(\cos y + i \sin y)$$

This definition is motivated by Euler's formula $e^{iy} = \cos y + i \sin y$.

3.1.1 Properties

1. **Differentiation:** The derivative of e^z is e^z everywhere. Thus, e^z is an *entire* function.

2. **Modulus and Argument:**

$$|e^z| = e^x \quad \text{and} \quad \arg(e^z) = y + 2n\pi \quad (n \in \mathbb{Z})$$

3. **Non-Zero:** $e^z \neq 0$ for any complex number z .

4. **Periodicity:** The function is periodic with a purely imaginary period of $2\pi i$:

$$e^{z+2\pi i} = e^z$$

In view of the periodicity property, we define the **fundamental strip** for the exponential function as $\mathbb{R} \times [0, 2\pi)$, which maps bijectively to $\mathbb{C} \setminus \{0\}$ under e^z , see Figure 3.1

3.2 The Logarithmic Function

Now we must define the *inverse* of the complex exponential function, i.e. the **complex logarithm**.

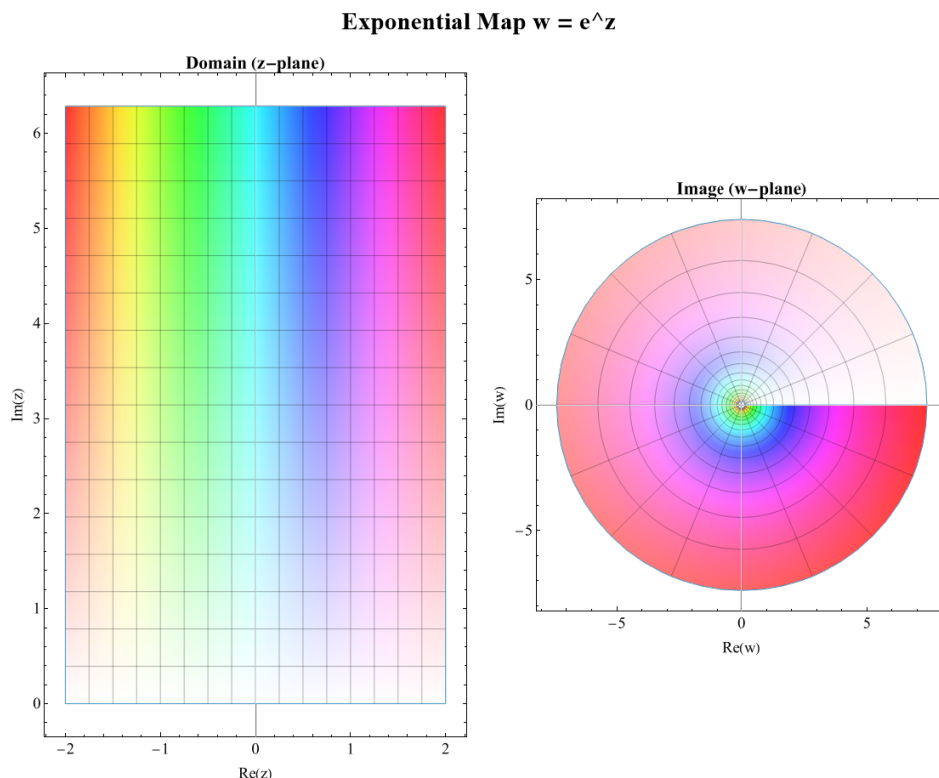


Figure 3.1: The mapping of the fundamental strip $\mathbb{R} \times [0, 2\pi)$ under the exponential function. Vertical lines map to circles, horizontal lines map to rays.

Definition 3.2.1 (Logarithm). For any nonzero complex number $z = re^{i\Theta}$ (where $\Theta = \text{Arg } z$), the logarithmic function is defined as:

$$\log z = \ln r + i(\Theta + 2n\pi) \quad (n \in \mathbb{Z})$$

or equivalently $\log z = \ln |z| + i \arg z$.

Definition 3.2.2 (Principal Value). The **Principal Value** of the logarithm, denoted by $\text{Log } z$, restricts the argument to the principal range $(-\pi, \pi]$:

$$\text{Log } z = \ln r + i\Theta \quad (-\pi < \Theta \leq \pi)$$

Remark 3.2.1. Notice that the $\text{Log}(z)$ is discontinuous along the negative real axis. Indeed for all $x \in \mathbb{R}, x > 0$ and $\varepsilon \in \mathbb{R}$

$$\lim_{\varepsilon \rightarrow 0^+} \text{Log}(-x + i\varepsilon) \neq \lim_{\varepsilon \rightarrow 0^+} \text{Log}(-x - i\varepsilon).$$

Exercise 3.2.1. Prove that for all $x > 0$

$$\lim_{\varepsilon \rightarrow 0^+} \text{Log}(-x + i\varepsilon) - \text{Log}(-x - i\varepsilon) = 2\pi i.$$

3.3 Branches and Derivatives of Logarithms

To treat the logarithm as a single-valued, differentiable function (suitable for calculus), we must restrict its domain.

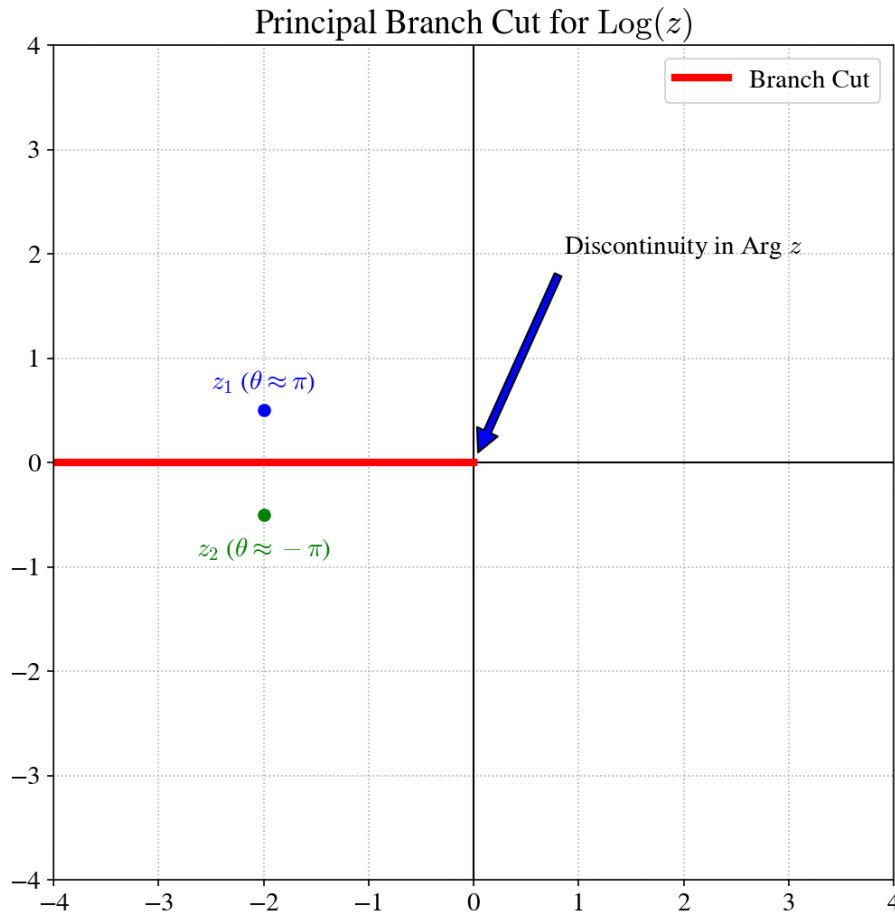


Figure 3.2: The branch-cut of the logarithm

Definition 3.3.1 (Branch of a Function). Let F be a single-valued function analytic in some domain D . If $F(z)$ is one of the values of a multiple-valued function $f(z)$ for every $z \in D$, then F is called a **branch** of f .

Definition 3.3.2 (Branch Cut). A **branch cut** is a portion of a line or curve that is introduced in order to define a branch F of a multiple-valued function f . Points on the branch cut are **singular points** (specifically, points of discontinuity) for that branch.

Example 3.3.1 (Principal Branch). The principal branch of the logarithm is:

$$\text{Log } z = \ln r + i\Theta \quad (r > 0, -\pi < \Theta < \pi)$$

Here, the branch cut is the **origin and the negative real axis** (where $\Theta = \pi$). The origin is called a **branch point**, see Fig 3.2

Theorem 3.3.1. For any branch of the logarithm defined on a domain D (not containing the origin), the derivative is:

$$\frac{d}{dz} \log z = \frac{1}{z}$$

3.3.1 Intermezzo: The Riemann Surface of the Logarithm

Introducing branch cuts is not the only way to make a multi-valued function single-valued. While a branch cut acts like a "brick wall" that stops us from walking completely around the origin and encountering multiple values, there is a much more elegant, deeply geometric approach: we can *change the space itself*.

Instead of restricting the function by cutting the plane, we can construct a new geometric domain called a **Riemann Surface**.

Imagine the flat complex plane (excluding the origin). As you walk around the origin in a circle, your angle θ continuously increases. On a normal, flat plane, after a full 360° (2π) rotation, you are standing on the exact same physical point where you started. This geometric collision is exactly what causes the logarithm to have multiple values at the same point!

But what if the complex plane wasn't flat? Imagine that as you walk counterclockwise around the origin, instead of returning to your starting point, you walk slightly upwards. The space behaves like an infinite spiral staircase or a circular parking garage ramp.

- After one full rotation (2π), you do not hit your starting point; you are on a new "sheet" directly above where you started.
- If you keep walking counterclockwise, you keep going up to higher and higher sheets.
- If you walk clockwise, your angle becomes negative, and you walk downwards onto lower sheets.

This infinite spiral is the Riemann surface for the complex logarithm.

Why is this so exciting? On this spiral surface, the point at angle 0 and the point at angle 2π are no longer the same point—they live on completely different levels of the staircase. Because every single continuous angle $\theta \in (-\infty, \infty)$ now has its own unique, physical location on the surface, the function $\log(z)$ only assigns **one value to one point**.

By allowing our 2D plane to stretch and warp into an infinite 3D spiral, we have completely "fixed" the logarithm! It becomes a perfectly smooth, single-valued, and continuous function everywhere on this new surface. This profound idea—solving algebraic problems (multi-valuedness) by inventing new geometric spaces—was introduced by Bernhard Riemann in 1851 and is the gateway to some of the most beautiful advanced mathematics, including Algebraic Geometry and Topology.

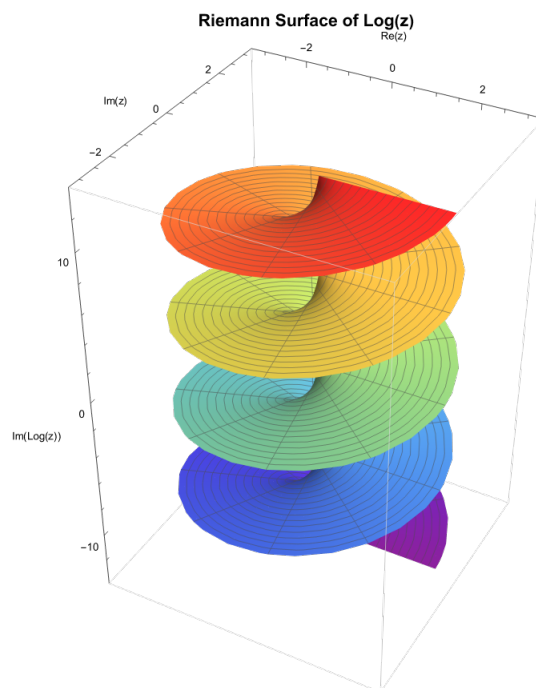
3.4 Trigonometric Functions

We define complex trigonometric functions using the exponential function, extending the Euler relations valid for real variables.

Definition 3.4.1 (Sine and Cosine).

$$\sin z = \frac{e^{iz} - e^{-iz}}{2i}, \quad \cos z = \frac{e^{iz} + e^{-iz}}{2}$$

Other functions ($\tan z$, $\cot z$, $\sec z$, $\csc z$) are defined in terms of sine and cosine in the usual way.

Figure 3.3: The infinite spiral of the $\text{Log}(z)$

3.4.1 Properties

1. **Derivatives:**

$$\frac{d}{dz} \sin z = \cos z, \quad \frac{d}{dz} \cos z = -\sin z$$

2. **Identities:** The Pythagorean identity holds:

$$\sin^2 z + \cos^2 z = 1$$

3. **Zeros:**

- $\sin z = 0$ if and only if $z = n\pi$ (real zeros only).
- $\cos z = 0$ if and only if $z = \pi/2 + n\pi$ (real zeros only).

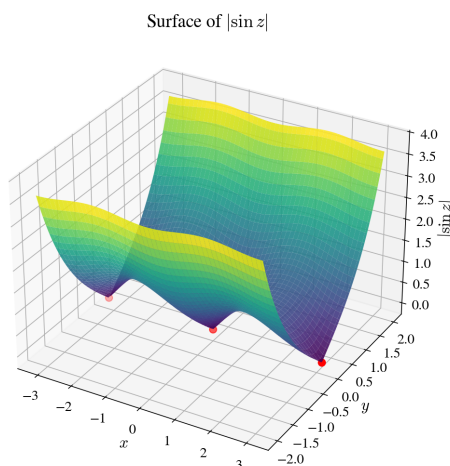
4. **Unboundedness:** Unlike real sine and cosine (bounded by 1), complex sine and cosine are **unbounded**. For example, on the imaginary axis $z = iy$:

$$|\sin(iy)| = \left| \frac{e^{-y} - e^y}{2i} \right| = \sinh y \rightarrow \infty \text{ as } y \rightarrow \infty$$

3.5 Hyperbolic Functions

Definition 3.5.1 (Sinh and Cosh). The hyperbolic sine and cosine are defined as:

$$\sinh z = \frac{e^z - e^{-z}}{2}, \quad \cosh z = \frac{e^z + e^{-z}}{2}$$

Figure 3.4: $|\sin(z)|$

3.5.1 Relations to Trigonometric Functions

The hyperbolic and trigonometric functions are intimately connected by the imaginary unit i :

$$\begin{aligned}\sinh(iz) &= i \sin z \\ \cosh(iz) &= \cos z \\ \sin(iz) &= i \sinh z \\ \cos(iz) &= \cosh z\end{aligned}$$

Theorem 3.5.1 (Identities).

$$\cosh^2 z - \sinh^2 z = 1$$

3.6 Complex Exponents

Now that we have defined the complex logarithm, we can extend the concept of exponents to complex numbers.

Definition 3.6.1 (Complex Powers). When $z \neq 0$ and c is any complex number, the function z^c is defined by means of the equation:

$$z^c = e^{c \log z}$$

Since $\log z$ is a multiple-valued function, the complex power z^c is also generally multiple-valued. The number of distinct values depends entirely on the nature of the exponent c .

Definition 3.6.2 (Principal Value of z^c). If $z \neq 0$, the **principal value** of z^c is obtained by using the principal branch of the logarithm:

$$z^c = e^{c \operatorname{Log} z}$$

3.6.1 Derivatives and Branch Cuts

Any branch of z^c is analytic where the chosen branch of $\log z$ is analytic. The branch cut for z^c will be identical to the branch cut of the logarithm used to define it. Using the chain rule, the derivative of a chosen branch of z^c is:

$$\frac{d}{dz} z^c = \frac{d}{dz} e^{c \log z} = e^{c \log z} \cdot \frac{c}{z} = cz^{c-1}$$

Example 3.6.1 (Computing i^{-i}). Let us find the principal value of i^{-i} . Using the definition:

$$i^{-i} = \exp(-i \operatorname{Log} i)$$

Since $|i| = 1$ and the principal argument is $\operatorname{Arg} i = \pi/2$, we have $\operatorname{Log} i = \ln(1) + i(\pi/2) = i\pi/2$. Therefore:

$$i^{-i} = \exp\left(-i \left(i \frac{\pi}{2}\right)\right) = \exp\left(\frac{\pi}{2}\right) = e^{\pi/2}$$

Notice that the principal value of i^{-i} is a purely real number!

3.6.2 A Deeper Look at Branch Cuts: Square Roots

It is tempting to manipulate complex exponents using familiar real-number algebra, such as writing $(ab)^c = a^c b^c$. However, because of branch cuts, this can fail spectacularly in the complex plane.

Let us analyze two functions that appear algebraically identical but have very different branch cuts when defined using the principal branch of the logarithm.

Function 1: $S(z) = (z^2 - 1)^{1/2}$ By definition, $S(z) = \exp\left(\frac{1}{2} \operatorname{Log}(z^2 - 1)\right)$. The principal logarithm has a branch cut whenever its argument is a negative real number or zero. Thus, $S(z)$ has a cut whenever $z^2 - 1 \in (-\infty, 0]$. Let $z = x + iy$. Then $z^2 - 1 = (x^2 - y^2 - 1) + i(2xy)$. For this to be real and non-positive, we need:

$$1. \ 2xy = 0 \implies x = 0 \text{ or } y = 0.$$

$$2. \ x^2 - y^2 - 1 \leq 0.$$

If $y = 0$ (the real axis), the condition becomes $x^2 - 1 \leq 0 \implies x \in [-1, 1]$.

If $x = 0$ (the imaginary axis), the condition becomes $-y^2 - 1 \leq 0$, which is true for *all* y . Therefore, the branch cut for $S(z)$ forms a cross: the line segment $[-1, 1]$ on the real axis, *and* the entire imaginary axis!

Function 2: $R(z) = (z-1)^{1/2}(z+1)^{1/2}$ By definition, $R(z) = \exp\left(\frac{1}{2} \operatorname{Log}(z-1)\right) \exp\left(\frac{1}{2} \operatorname{Log}(z+1)\right)$.

- The first term has a branch cut where $z-1 \in (-\infty, 0]$, which is the ray $x \leq 1, y = 0$.
- The second term has a branch cut where $z+1 \in (-\infty, 0]$, which is the ray $x \leq -1, y = 0$.

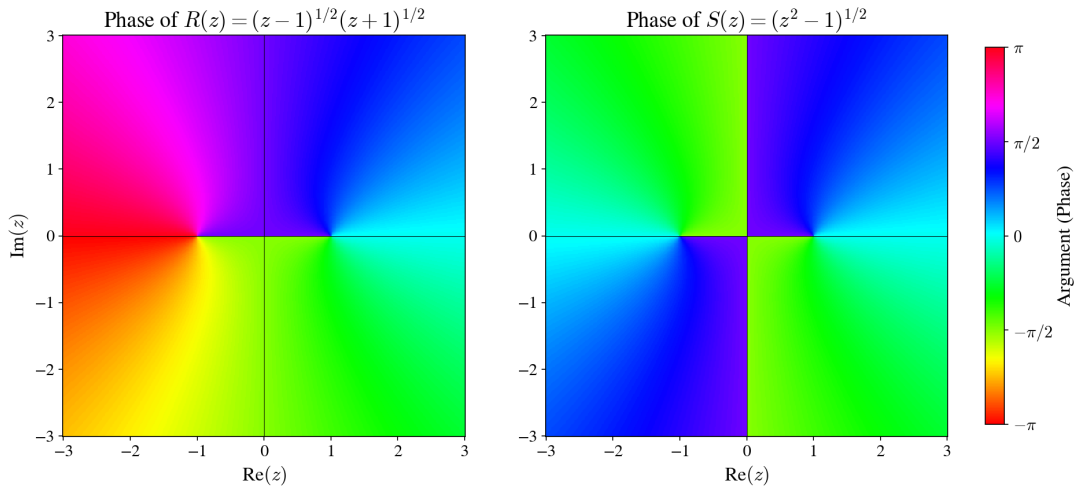


Figure 3.5: Comparison between the branch-cuts of $R(z) = (z-1)^{1/2}(z+1)^{1/2}$ and $S(z) = (z^2-1)^{1/2}$

It seems the cut for $R(z)$ is the entire ray $(-\infty, 1]$. However, let's look at the segment $(-\infty, -1)$ where both cuts overlap. If we approach the real axis from above ($y \rightarrow 0^+$), both $(z-1)$ and $(z+1)$ have an angle of π . The product evaluates to $e^{i\pi/2}e^{i\pi/2} = e^{i\pi} = -1$. If we approach from below ($y \rightarrow 0^-$), both terms have an angle of $-\pi$. The product evaluates to $e^{-i\pi/2}e^{-i\pi/2} = e^{-i\pi} = -1$. Because the limits from above and below agree, the discontinuity *cancels out*! Therefore, the branch cut for $R(z)$ is only the line segment $[-1, 1]$.

3.7 Exercises for Chapter 3

Exercise 3.7.1. Show that $\exp(z + \pi i) = -\exp z$.

Exercise 3.7.2. Find all values of z such that $e^z = -2$. (*Hint: Use the Logarithm*).

Exercise 3.7.3. Prove that $|\sin z|^2 = \sin^2 x + \sinh^2 y$. Use this to verify that $|\sin z|$ is unbounded.

Exercise 3.7.4. Compute the principal value $\text{Log}(-1 + i)$.

Exercise 3.7.5. Show that the function $f(z) = \text{Log}(z - i)$ has a branch point at $z = i$. Sketch a possible branch cut.

Exercise 3.7.6 (Values of Complex Powers). Using the definition $z^c = e^{c \log z}$, compute all possible values for the following complex powers. For each, explicitly identify the **principal value** (obtained by using the principal branch of the logarithm, $\text{Log}(z)$).

- (a) $1^{\sqrt{2}}$
- (b) $(-i)^i$
- (c) $(1 + i)^{1-i}$

Exercise 3.7.7 (Branch Cuts of Complex Powers). Let $f(z) = (z + 2i)^i$.

- (a) Write the explicit definition of the *principal branch* of this function.
- (b) Determine the exact location of the branch cut for this principal branch. Which ray in the complex plane must be removed to make the function analytic? Sketch it.
- (c) Compute the derivative $f'(z)$ for this branch. Where is this derivative well-defined?

Exercise 3.7.8 (Analyzing Branch Cuts of Algebraic Functions). Consider the function $g(z)$ defined using the principal branch of the complex logarithm:

$$g(z) = \exp\left(\frac{1}{2}\text{Log}(z^2 + 1)\right)$$

- (a) For what values of w does $\text{Log}(w)$ have its branch cut?
- (b) Set $w = z^2 + 1$. By letting $z = x + iy$, find the exact set of points (x, y) in the complex plane where $z^2 + 1$ falls on the branch cut of the principal logarithm.
- (c) Sketch the resulting branch cuts for $g(z)$.

Hint: The cuts should lie on the imaginary axis. Find their exact endpoints.

Exercise 3.7.9 (Overlapping and Canceling Branch Cuts). Consider the multi-valued function $h(z) = z^{1/2}(z - 2)^{1/2}$. Suppose we define a specific branch of $h(z)$ by taking the principal branch of the square root for both terms:

$$H(z) = \exp\left(\frac{1}{2}\text{Log}(z)\right) \exp\left(\frac{1}{2}\text{Log}(z - 2)\right)$$

- (a) Identify the individual branch cuts for $\exp\left(\frac{1}{2}\text{Log}(z)\right)$ and $\exp\left(\frac{1}{2}\text{Log}(z - 2)\right)$.
- (b) Consider the segment of the negative real axis $(-\infty, 0)$. This segment lies on the branch cut for *both* terms. Pick a point $x_0 < 0$ on this segment. Calculate the limit of $H(z)$ as $z \rightarrow x_0$ from the upper half-plane ($y \rightarrow 0^+$) and from the lower half-plane ($y \rightarrow 0^-$).
- (c) Show that these two limits are exactly the same. Conclude that the branch cut “cancels out” on $(-\infty, 0)$ and that the true branch cut for $H(z)$ is strictly the line segment $[0, 2]$.

Chapter 4

Complex Integration

4.1 Complex Integration

Integration in the complex plane is intimately connected to paths or “contours”. Before diving into the deep topological consequences of complex integration, we establish the fundamental mechanics.

4.1.1 Contours and Contour Integrals

A **contour** C in the complex plane is a piecewise smooth curve given by a parameterization $z(t) = x(t) + iy(t)$ for $a \leq t \leq b$. The integral of a complex-valued function $f(z)$ along a contour C is defined as:

$$\int_C f(z) dz = \int_a^b f(z(t)) z'(t) dt$$

4.1.2 Antiderivatives and Independence of Path

Before discussing integrals over closed loops, it is crucial to understand when an integral depends *only* on its endpoints.

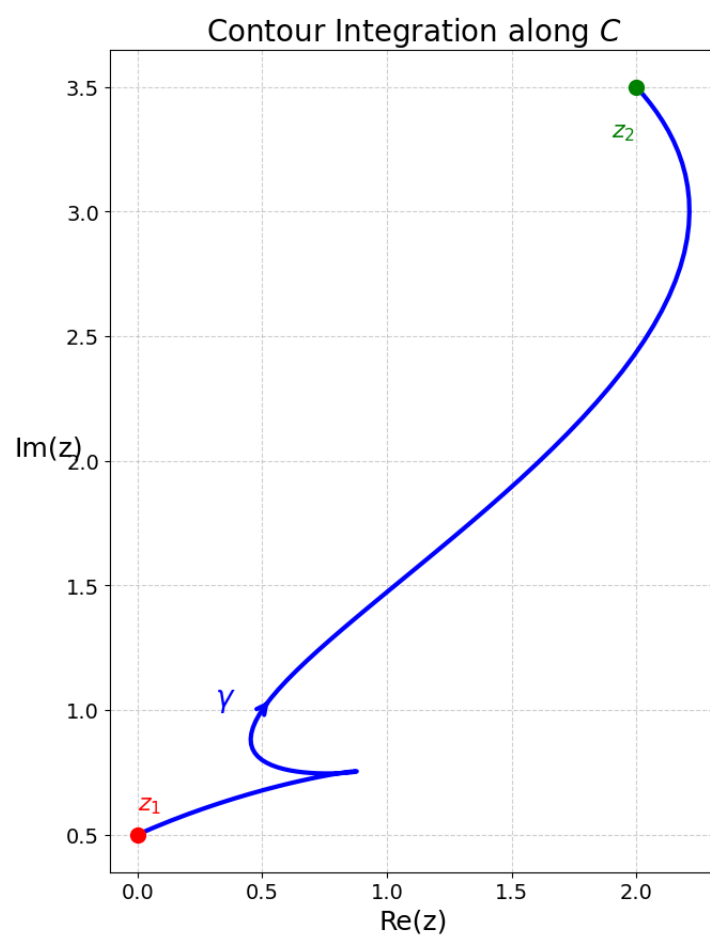
Definition 4.1.1 (Antiderivative). A function $F(z)$ is an antiderivative of $f(z)$ in a domain D if $F'(z) = f(z)$ for all $z \in D$. Note that this strictly requires $F(z)$ to be analytic in D .

Theorem 4.1.1 (Fundamental Theorem of Calculus for Contour Integrals). *Suppose that a function $f(z)$ is continuous on a domain D and has an antiderivative $F(z)$ throughout D . Then, for any contour C lying entirely in D with initial point z_1 and final point z_2 :*

$$\int_C f(z) dz = F(z_2) - F(z_1)$$

Consequences:

1. **Path Independence:** If f has an antiderivative, the integral between z_1 and z_2 is identical for *any* contour connecting them within D .

Figure 4.1: An example of a contour γ

2. **Closed Contours:** If C is a closed contour ($z_1 = z_2$), then:

$$\oint_C f(z) dz = F(z_1) - F(z_1) = 0$$

Example 4.1.1. Consider $\int_C z^2 dz$ where C is the upper half of the unit circle from $z = 1$ to $z = -1$. Since z^2 is entire, it has the antiderivative $F(z) = \frac{z^3}{3}$ everywhere. Thus:

$$\int_C z^2 dz = \frac{(-1)^3}{3} - \frac{1^3}{3} = -\frac{2}{3}$$

4.1.3 The Cauchy-Goursat Theorem

The fact that closed contour integrals of analytic functions evaluate to zero is the cornerstone of complex analysis. We will first prove this using a familiar tool from multivariable calculus, and then generalize it.

Proof via Green's Theorem (Assuming f' is continuous)

Historically, Augustin-Louis Cauchy proved his famous theorem with the additional assumption that the derivative $f'(z)$ is continuous. Under this assumption, the result follows immediately from Green's Theorem.

Let $f(z) = u(x, y) + iv(x, y)$ be analytic in a simply connected domain D , and let C be a simple closed contour in D . We can write $dz = dx + idy$. Expanding the integral:

$$\begin{aligned} \oint_C f(z) dz &= \oint_C (u + iv)(dx + idy) \\ &= \oint_C (u dx - v dy) + i \oint_C (v dx + u dy) \end{aligned}$$

Since we assumed $f'(z)$ is continuous, the partial derivatives of u and v are continuous. We can apply Green's Theorem ($\oint_C P dx + Q dy = \iint_R (Q_x - P_y) dA$) to both real line integrals, where R is the region enclosed by C :

$$\oint_C f(z) dz = \iint_R (-v_x - u_y) dA + i \iint_R (u_x - v_y) dA$$

Because $f(z)$ is analytic, u and v satisfy the Cauchy-Riemann equations: $u_x = v_y$ and $u_y = -v_x$. Substituting these into the double integrals yields:

$$\iint_R (u_y - u_y) dA + i \iint_R (u_x - u_x) dA = 0 + 0i = 0$$

This beautifully geometric proof relies heavily on the continuity of the partial derivatives.

Goursat's Generalization (Compact Sets)

Decades later, Édouard Goursat proved that the assumption of a continuous derivative is completely unnecessary. Analyticity (the mere existence of the derivative) is sufficient!

Idea of the Proof: We introduce the concept of a **compact set** (a set that is both closed and bounded). The region R enclosed by our contour C is compact. A fundamental property of compact sets is that any open cover has a finite subcover.

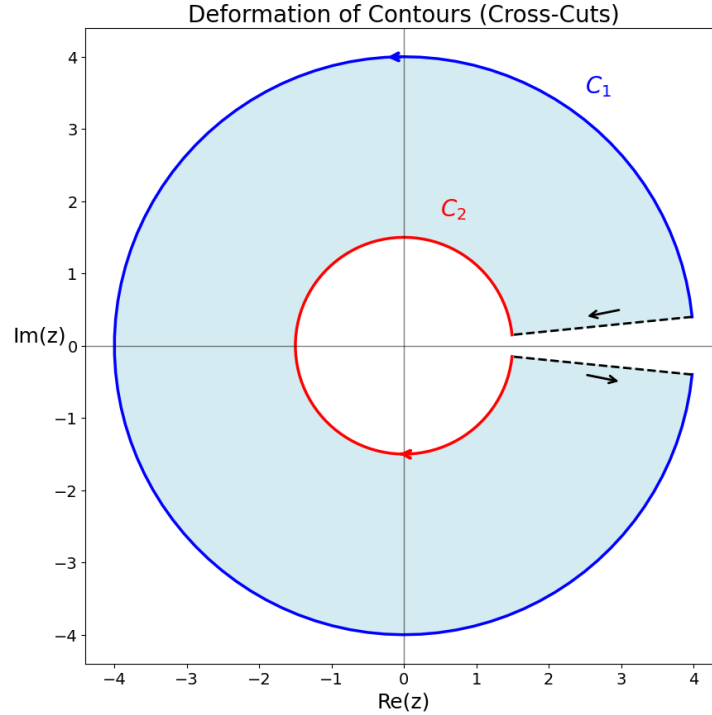


Figure 4.2: An application of Cauchy-Goursat: the integral over C_1 equals the integral over C_2 because the region between them contains no singularities.

If we subdivide the region R into a mesh of tiny squares and partial squares, the analyticity of $f(z)$ guarantees that locally, $f(z)$ is extremely close to its linear approximation: $f(z) \approx f(z_j) + f'(z_j)(z - z_j)$. Because R is compact, we can find a *finite* number of such small squares that cover the entire region such that the error of this linear approximation is uniformly bounded by some arbitrarily small ϵ .

When we integrate over the boundaries of all these tiny sub-squares, the interior boundaries cancel out completely (since each inner edge is traversed twice in opposite directions), leaving only the integral over the outer contour C . The integral over the linear parts ($f(z_j)$ and $f'(z_j)(z - z_j)$) evaluates exactly to zero because polynomials have antiderivatives. What remains is bounded by the arbitrarily small error ϵ multiplied by the area. Since ϵ can be chosen to be strictly greater than zero but arbitrarily small, the total integral must be exactly 0.

Theorem 4.1.2 (Cauchy-Goursat Theorem). *If a function f is analytic at all points interior to and on a simple closed contour C , then:*

$$\oint_C f(z) dz = 0$$

4.1.4 The Cauchy Integral Formula

If $f(z)$ is analytic inside a simple closed contour, its integral is zero. But what if there is a really specific singularity in the domain of the form $(z - z_0)$? This allows us to obtain the first representation theorem in complex analysis, the Cauchy Integral Formula, which expresses the value of an analytic function at a point in terms of an integral around a contour enclosing that point.

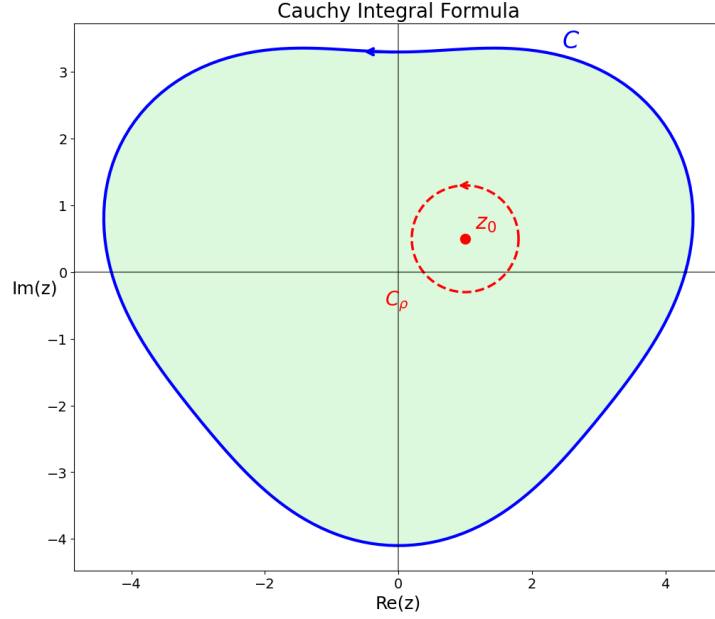


Figure 4.3: The Cauchy Integral Formula allows us to compute $f(z_0)$ by integrating $f(z)$ around a contour enclosing z_0 .

Theorem 4.1.3 (Cauchy Integral Formula). *Let f be analytic everywhere inside and on a simple closed contour C , taken in the positive (counterclockwise) sense. If z_0 is any point interior to C , then:*

$$f(z_0) = \frac{1}{2\pi i} \oint_C \frac{f(z)}{z - z_0} dz$$

Proof. Let C_ρ be a tiny circle of radius ρ centered at z_0 , fully contained inside C . By the Principle of Deformation of Contours, we can shrink C down to C_ρ :

$$\oint_C \frac{f(z)}{z - z_0} dz = \oint_{C_\rho} \frac{f(z)}{z - z_0} dz$$

Add and subtract $f(z_0)$ in the numerator:

$$\oint_C \frac{f(z)}{z - z_0} dz = \oint_{C_\rho} \frac{f(z_0)}{z - z_0} dz + \oint_{C_\rho} \frac{f(z) - f(z_0)}{z - z_0} dz$$

We know that $\oint_{C_\rho} \frac{1}{z - z_0} dz = 2\pi i$. For the second integral, because f is continuous at z_0 , $|f(z) - f(z_0)|$ becomes arbitrarily small as $\rho \rightarrow 0$. Since the length of C_ρ is $2\pi\rho$, the second integral vanishes in the limit. Thus:

$$\oint_C \frac{f(z)}{z - z_0} dz = f(z_0)(2\pi i) + 0$$

□

4.1.5 Extended Cauchy Integral Formula (Derivatives)

A stunning consequence of the Cauchy Integral Formula is that if a function is analytic (differentiable once), it is infinitely differentiable!

By formally differentiating the Cauchy Integral Formula with respect to z_0 under the integral sign, we obtain:

$$f'(z_0) = \frac{1}{2\pi i} \oint_C \frac{f(z)}{(z - z_0)^2} dz$$

Repeating this process n times yields the Generalized Cauchy Integral Formula:

Theorem 4.1.4. *Given a simple closed contour C , assume that $f(z)$ is analytic inside C , then for all z_0 inside C :*

$$f^{(n)}(z_0) = \frac{n!}{2\pi i} \oint_C \frac{f(z)}{(z - z_0)^{n+1}} dz.$$

4.2 The Magic of the Cauchy Integral Formula

In the previous sections, we discovered that if a function is analytic, it is infinitely differentiable. This is already a miracle! But Cauchy's Integral Formula is a gift that keeps on giving. By simply looking at the sizes (moduli) of these integrals, we can prove some of the most profound theorems in all of mathematics.

4.2.1 Cauchy's Inequality and Liouville's Theorem

Let's recall the extended Cauchy Integral Formula for the first derivative at a point z_0 , integrating around a circle C_R of radius R centered at z_0 :

$$f'(z_0) = \frac{1}{2\pi i} \oint_{C_R} \frac{f(z)}{(z - z_0)^2} dz$$

Suppose the function $f(z)$ is bounded on this circle, meaning there is some positive real number M such that $|f(z)| \leq M$ for all z on C_R . We can bound the size of the derivative using the standard ML -inequality (Maximum modulus $\times$ Length of contour):

$$|f'(z_0)| \leq \frac{1}{2\pi} \cdot \frac{M}{R^2} \cdot (2\pi R) = \frac{M}{R}$$

This beautifully simple bound is known as **Cauchy's Inequality**. Notice what happens if $f(z)$ is an *entire* function (analytic everywhere in $\mathbb{C}$). We can make the radius R as large as we want!

Theorem 4.2.1 (Liouville's Theorem). *If a function f is entire and bounded in the complex plane, then $f(z)$ is constant throughout the plane.*

Proof. Since f is bounded everywhere, there exists an M such that $|f(z)| \leq M$ for all $z \in \mathbb{C}$. Pick any point $z_0 \in \mathbb{C}$. By Cauchy's Inequality, $|f'(z_0)| \leq \frac{M}{R}$ for any radius R . Taking the limit as $R \rightarrow \infty$, the right side goes to 0. Thus, $|f'(z_0)| = 0 \implies f'(z_0) = 0$. Since the derivative is zero everywhere, $f(z)$ must be a constant function! $\square$

Why you should be surprised: Think about the real numbers. The function $g(x) = \sin(x)$ is defined everywhere on the real line and is perfectly bounded ($-1 \leq \sin(x) \leq 1$), yet it is clearly not constant. Liouville's Theorem tells us that the complex plane is far more rigid. If you try to make a complex function wavy but keep it bounded everywhere, analyticity breaks! (Indeed, we already know that $|\sin(z)|$ blows up to infinity along the imaginary axis).

Fundamental Theorem of Algebra: Roots of $P(z) = z^3 - 1$

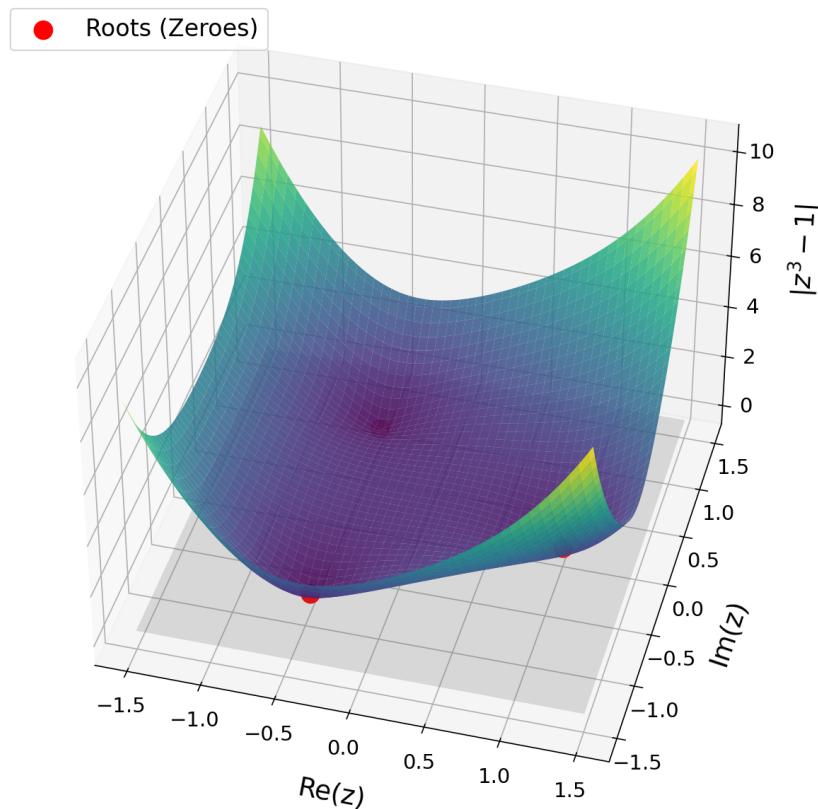


Figure 4.4: Example for the roots of a degree 3 polynomial

4.2.2 The Fundamental Theorem of Algebra

Liouville's Theorem gives us a shockingly easy proof of a theorem you've used since high school.

Theorem 4.2.2 (Fundamental Theorem of Algebra). *Any polynomial $P(z) = a_n z^n + a_{n-1} z^{n-1} + \cdots + a_1 z + a_0$ of degree $n \geq 1$ has at least one root in the complex plane.*

Proof. We proceed by contradiction. Suppose $P(z)$ has *no roots* anywhere in $\mathbb{C}$. If $P(z)$ is never zero, then the reciprocal function $f(z) = \frac{1}{P(z)}$ is entirely analytic (it's an entire function). Because $P(z)$ is a polynomial of degree ≥ 1 , $|P(z)| \rightarrow \infty$ as $|z| \rightarrow \infty$. This means that for sufficiently large R , $|f(z)| = \frac{1}{|P(z)|}$ is arbitrarily small. Inside the closed disk $|z| \leq R$, $f(z)$ is continuous and therefore bounded. So, $f(z)$ is bounded everywhere in $\mathbb{C}$. By Liouville's Theorem, $f(z)$ must be a constant. But if $1/P(z)$ is constant, then $P(z)$ is constant, which contradicts our assumption that $P(z)$ has degree $n \geq 1$. Thus, our assumption was wrong: $P(z)$ must have at least one root. $\square$

4.2.3 Gauss's Mean Value Theorem

Before we tackle the Maximum Modulus Principle, we need one more geometric insight from Cauchy's Integral Formula. If we write the path C_ρ as $z = z_0 + \rho e^{i\theta}$ for $0 \leq \theta \leq 2\pi$,

then $dz = i\rho e^{i\theta}d\theta$. Plugging this into the Cauchy Integral Formula for $f(z_0)$:

$$f(z_0) = \frac{1}{2\pi i} \int_0^{2\pi} \frac{f(z_0 + \rho e^{i\theta})}{\rho e^{i\theta}} i\rho e^{i\theta} d\theta = \frac{1}{2\pi} \int_0^{2\pi} f(z_0 + \rho e^{i\theta}) d\theta \quad (4.1)$$

This is **Gauss's Mean Value Theorem**. It says something beautiful: *The value of an analytic function at the center of a circle is exactly the average (mean) of its values on the perimeter of the circle.*

4.2.4 The Maximum Modulus Principle

We now dive into another important property of Complex functions, that has several applications in physics and engineering.

Theorem 4.2.3 (Maximum Modulus Principle). *If a function f is analytic and not constant in a given domain D , then its modulus $|f(z)|$ has no maximum value in D . That is, there is no point z_0 in the interior of D such that $|f(z)| \leq |f(z_0)|$ for all z around it.*

Proof. We proceed by contradiction, therefore we assume that $|f(z)|$ does have a local maximum at a point z_0 in D . This means there exists a circle C_ρ centered at z_0 such that for all z on and inside the circle, $|f(z)| \leq |f(z_0)|$.

From Gauss's Mean Value Formula (4.1), we know:

$$f(z_0) = \frac{1}{2\pi} \int_0^{2\pi} f(z_0 + \rho e^{i\theta}) d\theta$$

Taking the absolute value (modulus) of both sides, and using the fact that the absolute value of an integral is less than or equal to the integral of the absolute value:

$$|f(z_0)| \leq \frac{1}{2\pi} \int_0^{2\pi} |f(z_0 + \rho e^{i\theta})| d\theta$$

Now, recall our assumption: $|f(z_0 + \rho e^{i\theta})| \leq |f(z_0)|$ for all θ . If even for a tiny arc the strict inequality $|f(z_0 + \rho e^{i\theta})| < |f(z_0)|$ held true, the average value on the right side of Equation 4.2.4 would be strictly less than $|f(z_0)|$. But the inequality says $|f(z_0)| \leq \text{Average}$, which would create the contradiction $|f(z_0)| < |f(z_0)|$.

Therefore, $|f(z)|$ must be exactly equal to $|f(z_0)|$ everywhere on the circle C_ρ . Since we can repeat this for any smaller radius, $|f(z)|$ is constant throughout the entire disk around z_0 .

By a theorem from Chapter 2, if the modulus of an analytic function is constant in a domain, the function itself must be constant! This contradicts our premise that $f(z)$ is non-constant. Hence, no such maximum can exist in the interior. $\square$

Corollary 4.2.4. *If $f(z)$ is analytic and non-constant on a closed, bounded region R , the maximum value of $|f(z)|$ occurs **somewhere on the boundary** of R , and never in the interior.*

4.2.5 Physical Application: Laplace's Equation and Heat

In Chapter 2, we learned that the real and imaginary parts of an analytic function $f(z) = u(x, y) + iv(x, y)$ are **harmonic functions**. That means they satisfy Laplace's Equation:

$$\Delta u = u_{xx} + u_{yy} = 0$$

Maximum Principle: Harmonic Function $u(x, y) = x^2 - y^2$

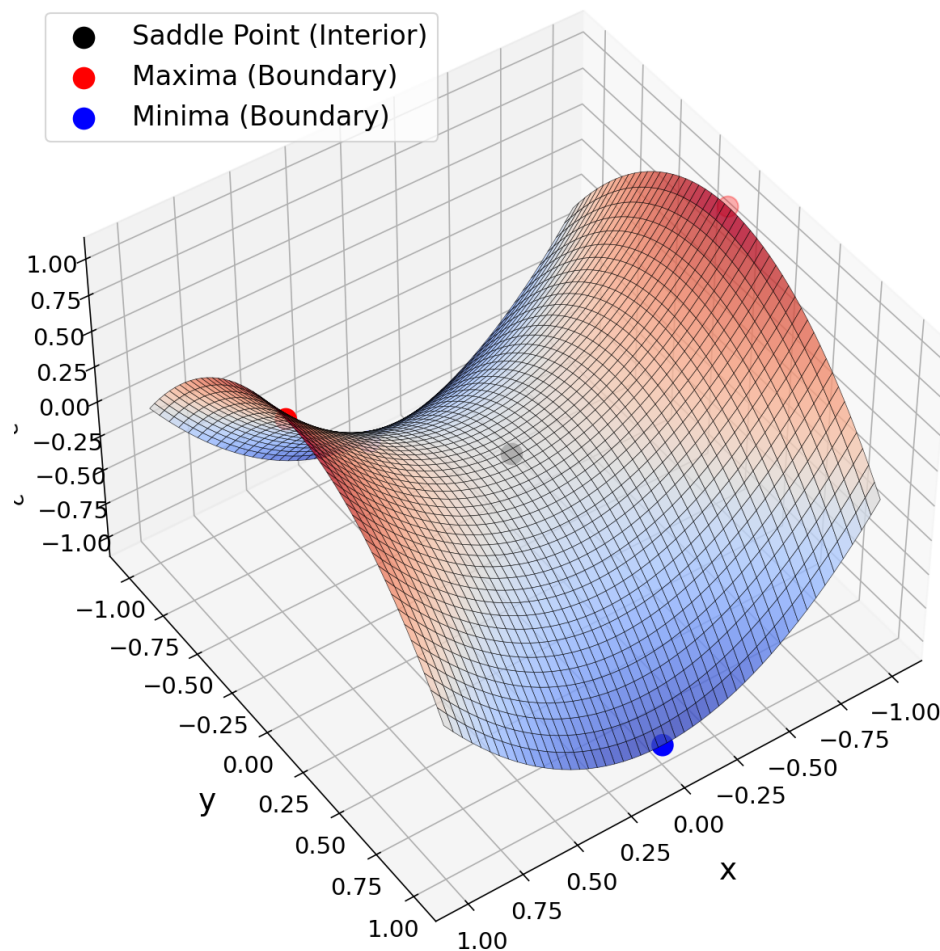


Figure 4.5: Example of Laplace equation

Laplace's equation models steady-state temperature distribution. If $u(x, y)$ represents the temperature of a metal plate, the Maximum Modulus Principle transfers directly to harmonic functions:

Maximum Principle for Harmonic Functions: *A non-constant harmonic function $u(x, y)$ cannot have a local maximum or minimum in the interior of its domain. The hottest and coldest points must lie on the boundary.*

Physical Intuition: If there were a localized “hottest point” in the middle of a metal plate, heat would spontaneously flow outward from that point, cooling it down. Therefore, the system wouldn't be in a “steady state.” The only way a point can stay hotter than its surroundings is if it is attached to a heat source on the boundary!

4.3 Exercises and Solutions

Exercise 4.3.1. Evaluate $\oint_C \frac{e^{iz}}{z^2 + 1} dz$ where C is the circle $|z - i| = 1$ traversed counterclockwise.

Solution: First, factor the denominator: $z^2 + 1 = (z - i)(z + i)$. The contour C is a circle of radius 1 centered at i . The singularity $z = i$ is inside C , while $z = -i$ is outside C . We rewrite the integrand to isolate the analytic part:

$$\oint_C \frac{\left(\frac{e^{iz}}{z+i}\right)}{z-i} dz$$

Let $f(z) = \frac{e^{iz}}{z+i}$. This function is analytic everywhere inside and on C . Applying the Cauchy Integral Formula with $z_0 = i$:

$$\oint_C \frac{f(z)}{z-i} dz = 2\pi i f(i)$$

Compute $f(i)$:

$$f(i) = \frac{e^{i(i)}}{i+i} = \frac{e^{-1}}{2i}$$

Substitute this back:

$$\text{Result} = 2\pi i \left(\frac{1}{2ie} \right) = \frac{\pi}{e}$$

Exercise 4.3.2. Evaluate $\int_C \frac{\cos(z)}{e^z - 1} dz$ where C is the circle $|z| = 1/2$ traversed counterclockwise.

Solution: The integrand has singularities where $e^z - 1 = 0 \implies e^z = 1 \implies z = 2k\pi i$ for integers k . The only singularity that falls inside the contour $|z| = 1/2$ is $z = 0$ (when $k = 0$). Wait, we cannot directly apply the Cauchy Integral Formula because the denominator is not $(z - 0)^n$. However, we can use a Maclaurin series expansion for the denominator to factor out z :

$$e^z - 1 = \left(1 + z + \frac{z^2}{2} + \dots \right) - 1 = z \left(1 + \frac{z}{2} + \dots \right)$$

So the integrand is:

$$\frac{\cos(z)}{z(1 + z/2 + \dots)}$$

Let $f(z) = \frac{\cos(z)}{1 + z/2 + \dots} = \frac{z \cos(z)}{e^z - 1}$ for $z \neq 0$, and define $f(0) = \lim_{z \rightarrow 0} f(z) = 1$. Now $f(z)$ is analytic inside $|z| = 1/2$. We rewrite the integral as:

$$\oint_C \frac{f(z)}{z} dz = 2\pi i f(0)$$

Since $f(0) = 1$, the integral evaluates to $2\pi i$.

Exercise 4.3.3. Evaluate $\oint_{|z|=3} \frac{z^3 + 2}{(z - 1)^3} dz$ traversed counterclockwise.

Solution: The singularity $z_0 = 1$ is inside the circle $|z| = 3$. The power in the denominator is 3, so we use the Extended Cauchy Integral Formula with $n = 2$:

$$\oint_C \frac{f(z)}{(z - z_0)^{n+1}} dz = \frac{2\pi i}{n!} f^{(n)}(z_0)$$

Here, $f(z) = z^3 + 2$. We need the second derivative evaluated at $z_0 = 1$:
 1. $f'(z) = 3z^2$
 2. $f''(z) = 6z$ 3. $f''(1) = 6$

Plug into the formula ($n = 2 \implies 2! = 2$):

$$\oint_{|z|=3} \frac{z^3 + 2}{(z - 1)^3} dz = \frac{2\pi i}{2} (6) = 6\pi i$$

Exercise 4.3.4. Evaluate $\int_0^i z \cos(z^2) dz$ over any contour from 0 to i .

Solution: Because $f(z) = z \cos(z^2)$ is entire, it has an antiderivative everywhere. We can find it by substitution or inspection:

$$F(z) = \frac{1}{2} \sin(z^2)$$

Verify: $F'(z) = \frac{1}{2} \cos(z^2) \cdot (2z) = z \cos(z^2)$. By the Fundamental Theorem of Calculus for contour integrals:

$$\int_0^i z \cos(z^2) dz = F(i) - F(0) = \frac{1}{2} \sin(i^2) - \frac{1}{2} \sin(0) = \frac{1}{2} \sin(-1) = -\frac{1}{2} \sin(1)$$

The path taken from 0 to i does not matter!

Exercise 4.3.5. Evaluate $\oint_{|z|=2} \frac{\text{Log}(z + 3)}{z^2} dz$ where Log is the principal branch.

Solution: First, determine the branch cut for $g(z) = \text{Log}(z + 3)$. The principal logarithm has a cut when its argument is a negative real number or zero. $z + 3 \leq 0 \implies z \leq -3$. So the branch cut is the ray $(-\infty, -3]$ on the real axis. The contour is $|z| = 2$. The branch cut $(-\infty, -3]$ lies entirely *outside* the contour $|z| = 2$. Therefore, $f(z) = \text{Log}(z + 3)$ is analytic everywhere inside and on the contour. We have a singularity at $z_0 = 0$ due to the z^2 in the denominator. Apply the Extended CIF with $n = 1$:

$$\oint_C \frac{f(z)}{z^2} dz = \frac{2\pi i}{1!} f'(0)$$

Compute the derivative:

$$f'(z) = \frac{1}{z + 3} \implies f'(0) = \frac{1}{3}$$

$$\text{Result} = 2\pi i \left(\frac{1}{3} \right) = \frac{2\pi i}{3}$$

Exercise 4.3.6 (Applications of Liouville's Theorem). (a) Let $f(z)$ be an entire function. Suppose that the real part of $f(z)$ is bounded above, meaning $u(x, y) \leq M$ for some constant M . By considering the auxiliary function $g(z) = e^{f(z)}$, prove that $f(z)$ must be a constant function.

- (b) Suppose $f(z)$ is an entire function and that $|f(z)| \leq 10\sqrt{|z|}$ for all z . Prove that $f(z)$ must be exactly the zero function, $f(z) \equiv 0$.

Hint: Use Cauchy's Inequality for $f'(z_0)$ on a circle of radius R , and let $R \rightarrow \infty$.

Exercise 4.3.7 (Gauss's Mean Value Theorem). Let $f(z) = z^2$.

- (a) Calculate $f(0)$ directly.
- (b) Verify Gauss's Mean Value Theorem by explicitly calculating the integral:

$$\frac{1}{2\pi} \int_0^{2\pi} f(re^{i\theta}) d\theta$$

for a circle of radius $r = 2$. Show that your result matches part (a).

Exercise 4.3.8 (Finding the Maximum on a Boundary). Let $f(z) = z^2 + 2z - 1$. We know by the Maximum Modulus Principle that the maximum of $|f(z)|$ on the closed unit disk $|z| \leq 1$ must occur somewhere on the boundary circle $|z| = 1$.

- (a) Parameterize the boundary as $z = e^{i\theta}$.
- (b) Write an expression for $|f(e^{i\theta})|^2$ strictly in terms of θ .
Hint: $|w|^2 = w\bar{w}$, and remember that $\overline{e^{i\theta}} = e^{-i\theta}$.
- (c) Use standard calculus on your real-valued function of θ to find the exact point(s) z on the unit circle where $|f(z)|$ achieves its absolute maximum, and find the value of that maximum.

Exercise 4.3.9 (The Minimum Modulus Principle). The Maximum Modulus Principle tells us about maximums, but what about minimums?

- (a) Suppose $f(z)$ is analytic and non-constant in a domain D , and crucially, suppose $f(z) \neq 0$ anywhere in D . By applying the Maximum Modulus Principle to the function $g(z) = \frac{1}{f(z)}$, prove that $|f(z)|$ cannot have a local *minimum* in the interior of D .
- (b) Give a counter-example showing why the condition " $f(z) \neq 0$ " is absolutely necessary. (i.e., provide a simple analytic function that *does* achieve a minimum modulus in the interior of its domain).

Exercise 4.3.10 (Physical Application: Heat Distribution). Suppose the temperature in a square room $-1 \leq x \leq 1, -1 \leq y \leq 1$ is modeled by the steady-state (harmonic) function $T(x, y) = x^2 - y^2 + 2x + 10$.

- (a) Verify that $T(x, y)$ is indeed harmonic by showing it satisfies Laplace's equation.
- (b) Without finding the exact maximum, explain using the Maximum Principle for Harmonic Functions why a person looking for the warmest spot in the room should only check the walls (the boundary) and never the center of the room.
- (c) Find the analytic function $f(z)$ such that $T(x, y) = \operatorname{Re}(f(z))$.

Chapter 5

Sequences and Series

5.1 Convergence of Sequences and Series

One of the main tools in real analysis is the Taylor series. In this chapter we develop the parallel theory for complex analysis and we will see, as always, the complex numbers make everything more beautiful. We start by introducing the concept of sequence and series in the complex plane.

5.1.1 Sequences

An infinite sequence of complex numbers is denoted by $z_1, z_2, \dots, z_n, \dots$ or simply $\{z_n\}$. We say that the sequence converges to a limit z if, for every $\epsilon > 0$, there exists an integer $n_0 > 0$ such that:

$$|z_n - z| < \epsilon \quad \text{whenever} \quad n > n_0$$

Geometrically, this means that eventually, all terms of the sequence fall inside an arbitrarily small ϵ -neighborhood around z .

Because $z_n = x_n + iy_n$ and $z = x + iy$, a beautiful and incredibly useful theorem emerges:

Theorem 5.1.1. *A sequence $z_n = x_n + iy_n$ converges to $z = x + iy$ if and only if the sequence of real parts x_n converges to x , and the sequence of imaginary parts y_n converges to y .*

In this way, everything you learned from real analysis you can use it in the complex setting.

5.1.2 Series

An infinite series is the sum of a sequence: $\sum_{n=1}^{\infty} z_n = z_1 + z_2 + \dots$. We define the N -th partial sum as $S_N = \sum_{n=1}^N z_n$. The infinite series converges to a number S if and only if the sequence of partial sums $\{S_N\}$ converges to S . If a series does not converge, it diverges.

Absolute Convergence: If the series of absolute values $\sum_{n=1}^{\infty} |z_n|$ converges, we say the original complex series converges *absolutely*. Absolute convergence always implies standard convergence.

5.2 Taylor Series

We now arrive at one of the most powerful theorems in complex analysis. In real calculus, we required a function to be infinitely differentiable to even *write down* a Taylor series, and even then, it wasn't guaranteed to converge to the function. In the complex plane, analyticity alone guarantees everything.

Theorem 5.2.1 (Taylor's Theorem). *Suppose that a function f is analytic throughout a disk $D_0 = \{z : |z - z_0| < R_0\}$. Then $f(z)$ has the power series representation:*

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n \quad \text{where} \quad a_n = \frac{f^{(n)}(z_0)}{n!}$$

Furthermore, this series converges to $f(z)$ for every point z inside D_0 .

(Note: When $z_0 = 0$, this is called a Maclaurin series).

5.2.1 Detailed Proof of Taylor's Theorem

Step 1: Setup and the Cauchy Integral Formula.

Let z be any fixed point inside the disk D_0 . Let $r = |z - z_0|$. Construct a simple closed circular contour C_0 centered at z_0 with a radius r_0 such that $r < r_0 < R_0$. Because f is analytic inside and on C_0 , and z is strictly inside C_0 , the Cauchy Integral Formula tells us:

$$f(z) = \frac{1}{2\pi i} \oint_{C_0} \frac{f(s)}{s - z} ds$$

where s represents the points on the contour C_0 .

Figure Placeholder: Taylor Theorem Geometry

Caption: (Python Plot - Suggested) Draw a large circle representing the boundary of D_0 (radius R_0 , dashed). Inside, draw a smaller solid circle C_0 (radius r_0). Mark the center z_0 . Mark a point s on C_0 . Mark a point z strictly inside C_0 . Draw vectors to show $|s - z_0| = r_0$ and $|z - z_0| = r$.

Step 2: Algebraic Manipulation of the Denominator.

We want to express $\frac{1}{s-z}$ in terms of $(z - z_0)$. We can rewrite the denominator by adding

and subtracting z_0 :

$$\frac{1}{s-z} = \frac{1}{(s-z_0) - (z-z_0)} = \frac{1}{(s-z_0)} \cdot \frac{1}{1 - \left(\frac{z-z_0}{s-z_0}\right)}$$

Let $c = \frac{z-z_0}{s-z_0}$. Notice that $|c| = \frac{r}{r_0} < 1$.

Step 3: The Finite Geometric Progression.

From basic algebra, we know the finite geometric sum identity:

$$\frac{1}{1-c} = 1 + c + c^2 + \cdots + c^{N-1} + \frac{c^N}{1-c}$$

Substitute $c = \frac{z-z_0}{s-z_0}$ into this identity, multiply the whole thing by $\frac{1}{s-z_0}$, and we get:

$$\frac{1}{s-z} = \sum_{n=0}^{N-1} \frac{(z-z_0)^n}{(s-z_0)^{n+1}} + \frac{(z-z_0)^N}{(s-z)(s-z_0)^N}$$

Step 4: Integration and the Extended CIF.

Now, substitute this expanded version of $\frac{1}{s-z}$ back into our Cauchy Integral Formula (Equation 5.2.1):

$$f(z) = \frac{1}{2\pi i} \oint_{C_0} f(s) \left[\sum_{n=0}^{N-1} \frac{(z-z_0)^n}{(s-z_0)^{n+1}} + \frac{(z-z_0)^N}{(s-z)(s-z_0)^N} \right] ds$$

Because the sum is finite, we can pull it outside the integral:

$$f(z) = \sum_{n=0}^{N-1} (z-z_0)^n \left[\frac{1}{2\pi i} \oint_{C_0} \frac{f(s)}{(s-z_0)^{n+1}} ds \right] + R_N(z)$$

where $R_N(z)$ is the remainder term:

$$R_N(z) = \frac{(z-z_0)^N}{2\pi i} \oint_{C_0} \frac{f(s)}{(s-z)(s-z_0)^N} ds$$

Look closely at the bracketed integral in the sum! By the Extended Cauchy Integral Formula, that integral is exactly $\frac{f^{(n)}(z_0)}{n!}$. So we have:

$$f(z) = \sum_{n=0}^{N-1} \frac{f^{(n)}(z_0)}{n!} (z-z_0)^n + R_N(z)$$

Step 5: Bounding the Remainder.

To prove the infinite series converges to $f(z)$, we must show that $\lim_{N \rightarrow \infty} R_N(z) = 0$. Since $f(s)$ is continuous on the compact contour C_0 , it is bounded: $|f(s)| \leq M$. The distance from s to z is bounded below: $|s-z| \geq |s-z_0| - |z-z_0| = r_0 - r$. The length of C_0 is $2\pi r_0$. Using the ML -inequality on $R_N(z)$:

$$|R_N(z)| \leq \frac{r^N}{2\pi} \cdot \frac{M}{(r_0-r)r_0^N} \cdot 2\pi r_0 = \frac{Mr_0}{r_0-r} \left(\frac{r}{r_0}\right)^N$$

Since $r < r_0$, the ratio $(r/r_0) < 1$. Therefore, as $N \rightarrow \infty$, the term $(r/r_0)^N \rightarrow 0$. This proves that $R_N(z) \rightarrow 0$, and the Taylor series converges exactly to $f(z)$. $\square$

5.2.2 Examples and the Uniqueness of Series

Because analytic functions have unique power series representations, we rarely compute $a_n = \frac{f^{(n)}(z_0)}{n!}$ by taking endless derivatives. Instead, we algebraically manipulate known series.

Example: Find the Maclaurin series for $f(z) = z^2 e^{3z}$. We know that $e^w = \sum_{n=0}^{\infty} \frac{w^n}{n!}$ for all $w \in \mathbb{C}$. Substitute $w = 3z$:

$$e^{3z} = \sum_{n=0}^{\infty} \frac{(3z)^n}{n!} = \sum_{n=0}^{\infty} \frac{3^n}{n!} z^n$$

Now multiply by z^2 :

$$z^2 e^{3z} = z^2 \sum_{n=0}^{\infty} \frac{3^n}{n!} z^n = \sum_{n=0}^{\infty} \frac{3^n}{n!} z^{n+2}$$

This series converges for all $z \in \mathbb{C}$.

5.3 Exercises: Computing Series

Exercise 5.3.1. Using the known Maclaurin series for $\sin(z)$, find the Maclaurin series for the function $f(z) = \frac{\sin(z^2)}{z}$. (Assume $f(0) = 0$ so the function is entire). What is the radius of convergence?

Exercise 5.3.2. Find the Maclaurin series for $f(z) = \frac{1}{1+4z^2}$. State the exact region of the complex plane where this series converges.

Hint: Use the geometric series $\frac{1}{1-w} = \sum_{n=0}^{\infty} w^n$.

Exercise 5.3.3. Find the Taylor series for $f(z) = \frac{1}{z}$ centered at $z_0 = 1$.

Hint: Write $\frac{1}{z} = \frac{1}{1+(z-1)}$ and use the geometric series.

Exercise 5.3.4. By differentiating the Maclaurin series for $\frac{1}{1-z}$, find the Maclaurin series for $f(z) = \frac{1}{(1-z)^2}$.